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Nonlinear Interaction in the Characteristics of Real Assets

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Résumé

Ce projet vérifie si des dynamiques non linéaires et une dépendance à l'état macroéconomique, encore peu documentées pour les actifs réels, constituent des déterminants importants à leurs rendements. Alors que la littérature récente traite ces enjeux pour les actions, les actifs réels restent à la traîne, leurs caractéristiques tangibles et leurs structures de financement les rendant particulièrement prône à des interactions complexes. Dans un cadre séquentiel de régressions en panel (MCO), nous analysons des données trimestrielles sur les REIT américains et des données mensuelles sur les matières premières. La méthodologie évalue l'apport de transformations non linéaires, dont des termes polynomiaux cubiques, ainsi que d'interactions entre caractéristiques d'actifs et composantes principales macroéconomiques. Les modèles non linéaires et dépendants de l'état expliquent significativement mieux les rendements que les modèles linéaires de référence, avec près d'un doublement du pouvoir explicatif pour les REIT. Nous mettons en évidence un « corridor de valorisation » non monotone pour les rendements du FFO des REIT et montrons que les primes liées aux caractéristiques des REIT varient fortement selon le contexte, notamment une pénalité du levier qui s'accroît en période d'incertitude macroéconomique élevée. Pour les matières premières, l'effet négatif de la volatilité dépend de l'état, et le momentum est pénalisé en phase de forte croissance. Ces résultats appuient une approche plus dynamique de la valorisation et indiquent que les primes de risque associées aux caractéristiques des actifs réels ne sont pas statiques.

Mots clés : actifs réels, REIT, matières premières, évaluation d'actifs, effets non linéaires, dépendance à l'état, composantes principales

Méthodes de recherche : MCO en panel, analyse en composantes principales (ACP), termes polynomiaux, modèles avec interactions

Abstract

This project challenges the linear, additive assumptions of traditional asset pricing by investigating whether nonlinear dynamics and macroeconomic state-dependence are critical, underexplored drivers of real asset returns. While recent literature has embraced such models for equities, this gap remains for real assets, whose tangible characteristics and financing structures make them uniquely prone to complex interactions. Using a sequential panel OLS framework, this study analyzes quarterly U.S. REIT data and monthly commodity data. The methodology tests the incremental explanatory power of nonlinear transformations, such as cubic polynomial terms, and interactions between asset traits and macroeconomic principal components. The results show that nonlinear, state-dependent models materially outperform linear benchmarks, nearly doubling the explanatory power for REITs. We identify a robust, non-monotonic “valuation corridor” for REIT FFO yields and find that REIT trait premia are highly dynamic, most notably that the penalty for leverage intensifies during periods of high macro uncertainty. For commodities, we find that the negative impact of volatility is state-dependent, while momentum strategies are penalized during strong growth phases. These findings provide a strong empirical foundation for a more dynamic approach to valuation, challenging the sufficiency of linear models and demonstrating that the risk premia for real asset characteristics are not static.

Keywords: real assets, REITs, commodities, asset pricing, nonlinear effects, state dependence, principal components

Research methods: panel OLS, principal component analysis (PCA), polynomial terms, interaction models

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1. Introduction

Background & Motivation

Real assets, such as real estate and commodities, represent a significant component of institutional portfolios and are increasingly present in diversified retail portfolios. They are valued for their potential diversification benefits, inflation-hedging properties, and direct links to the real economy. Unlike equities or bonds, their pricing dynamics are heavily influenced by asset-specific characteristics such as leverage, depreciation, and cash flow stability.

Traditional empirical asset pricing frameworks typically assume that these characteristics contribute in a linear, additive manner. For example, the famous Fama-French factor model explains returns as a linear combination of exposures to systematic risk factors, which are themselves derived from firm characteristics such as size and book-to-market (Fama & French, 1993). In this framework, the contribution of a firm's "value" characteristic is assumed to be additive and independent of its "size" characteristic. However, this assumption of simple additivity may be an oversimplification. Financial theory and market experience suggest that these traits often interact or contribute to asset returns in nonlinear ways. For instance, leverage amplifies exposure to shocks, but the severity of the effect may depend critically on the stability of the underlying cash flows. Early research on REITs demonstrated that highly leveraged firms are significantly more sensitive to macroeconomic risk factors than their low-leverage peers (Chan et al., 1990).

Real assets, as an asset class, are particularly prone to nonlinearities. Commodities, for example, exhibit regime-sensitive demand, with assets like gold attracting significant investment during economic downturns. Other real assets, such as REITs, have heterogeneous financing structures and tangible cash flows, creating an environment susceptible to interaction effects. The 2008 Financial Crisis offered a striking illustration of this: highly leveraged REITs, particularly those with shorter debt maturities, experienced disproportionately large price declines compared to more conservatively financed peers (Sun et al., 2015), an outcome that linear models might struggle to capture.

Problem Statement

While advances in equity asset pricing emphasize the importance of nonlinearities and interaction effects (e.g., Dittmar, 2002; Gu et al., 2020; Kirby, 2020), this same line of questioning remains under-explored for real assets. The existing literature is still dominated by models that assume asset traits act independently and linearly (e.g., Coën & Guardiola, 2025; Yiu et al., 2022). This gap in the literature motivates the central questions of this project: Do combinations of real asset characteristics create nonlinear pricing dynamics that linear models miss? And if so, how significant are these effects for model fit and the economic interpretation of risk exposures?

Research Objectives

This project investigates whether nonlinear interactions between real asset characteristics improve return predictability and explanatory power compared to additive models. The analysis proceeds in several stages. First, we construct a dataset of key characteristics for real assets, focusing on REITs and commodities. These two asset classes provide complementary perspectives, with REITs representing income-producing, levered real assets, while commodities reflect physical goods traded in global markets. Second, we establish baseline linear models of returns on these characteristics. The core of the analysis then introduces interaction terms and nonlinear transformations to capture the hypothesized nonlinear effects. Finally, we evaluate whether these specifications improve model fit and better explain real asset returns.

This project contributes in three ways.

- Empirically, it extends nonlinear asset pricing insights to the important yet underexplored domain of real assets.
- Methodologically, it directly compares additive and nonlinear regression models in a unified econometric framework, testing the incremental value of these nonlinearities and interactions.
- Economically, it sheds light on how asset characteristics interact to shape risk exposures, offering insights relevant for portfolio construction and risk management.

Ultimately, this project seeks to determine whether accounting for nonlinear interactions between real asset characteristics materially improves our understanding of what drives their returns.

2. Literature Review

Introduction and Scope

Research on asset pricing has increasingly challenged the linear, additive assumptions that underpin many empirical models. A growing body of evidence shows that asset returns often reflect nonlinear and state-dependent relationships that standard models fail to capture. While these developments have reshaped our understanding of return predictability, they remain largely confined to financial assets such as equities and bonds.

The literature review below spans five themes. First, studies on nonlinear pricing kernels and higher moments (Almeida et al., 2022; Almeida & Garcia, 2017; Dittmar, 2002) show how nonlinear transformations of the stochastic discount factor improve explanatory power relative to traditional linear models. Second, papers on state dependence and regime switching (Caggiano et al., 2017; Mijatović & Schneider, 2014; Silvestrini & Zaghini, 2015) highlight how macroeconomic shocks have asymmetric effects across economic regimes. Third, nonlinear factor and characteristic models (Almeida & Freire, 2023; Kirby, 2020) demonstrate that the cross-sectional pricing of characteristics is not constant, often depending on other traits in ways that simple linear models cannot capture. Fourth, recent machine learning approaches (Chen et al., 2024; Gu et al., 2020) showcase flexible nonlinear estimators that uncover hidden interactions among predictors. Finally, behavioural and theoretical frameworks (Hirano & Toda, 2024; Schmitt et al., 2020) emphasize how nonlinear pricing dynamics can arise from the market's own internal structure, rather than just from external shocks.

Together, these themes indicate that nonlinearities are a key feature of financial markets. Yet little is known about whether similar mechanisms operate in real assets, such as REITs and commodities, whose tangible characteristics and financing structures make them particularly prone to nonlinear dynamics. The following sections synthesize this literature and establish the conceptual foundations for the empirical analysis that follows.

Nonlinear Pricing Kernels and Higher Moments

Prior research on nonlinear pricing kernels established that investor preferences for higher-order moments can materially influence expected returns. Dittmar (2002) develops a cubic expansion of aggregate wealth to model the stochastic discount factor (SDF), allowing investor utility

preferences for skewness and kurtosis. The nonlinear kernel is estimated using GMM on U.S. equity portfolios and explains the cross-section of returns significantly better than both the CAPM and Fama-French models. Once the cubic pricing kernel is augmented with Fama-French factors, those traditional factors lose statistical significance, indicating that higher-moment risk preferences capture much of what those factors proxy. Conceptually, Dittmar shows that polynomial expansions of wealth offer a tractable way to introduce nonlinearity as well as capture utility and asset payoffs beyond simple mean–variance.

Building on this foundation, Almeida & Garcia (2017) generalize the evaluation of SDFs by introducing nonparametric higher-moment bounds. Their framework extends the classic Hansen-Jagannathan variance-based approach by allowing a more precise decomposition of pricing kernel dispersion into skewness and kurtosis components. Applied to equity and option returns, these generalized minimum discrepancy bounds show that many traditional linear models are adequate when evaluated with variance but fail once higher-moment constraints are imposed. In particular, disaster risk and long-run risk models struggle to satisfy these tighter bounds, suggesting that pricing errors often result from neglected asymmetry or tail risk preferences. Applied to real assets, this result suggests that nonlinear interactions (for example, leverage $\times$ volatility) may play a similar role in shaping results when distributions are asymmetric or fat-tailed.

More recently, Almeida, Garcia & N'Dri (2022) propose a hybrid approach that integrates linear and nonlinear principal components (LPCs and NLPCs) to extract independent nonlinear factors. They use Brenier-map NLPCs to construct a hybrid SDF that blends both factor types. Empirically, the hybrid SDF results in higher out-of-sample R^2 and Sharpe ratios, with fewer factors, than purely linear approaches, particularly for “extreme” portfolios that standard models struggle to price. These results emphasize that linear factor models overlook important nonlinear dependencies embedded in return data and that incorporating nonlinear transformations can materially improve pricing models by uncovering hidden structures.

State Dependence and Regime Switching

While nonlinear pricing kernels emphasize how investor preferences generate higher-order effects, other research has shown that similar nonlinearities emerge from macroeconomic and market dynamics, where the propagation of shocks depends on the current economic regime.

Silvestrini and Zaghini (2015) develop a time-varying parameter VAR with stochastic volatility, estimated with Bayesian methods, to study how financial shocks transmit to the real economy. Their results show clear state dependence: in stable periods, financial shocks have limited impact on GDP and inflation, but during crises they amplify downturns, depress credit, and prolong recovery. The model captures this state dependence by allowing both coefficients and volatilities to evolve over time. The key takeaway is that linear frameworks systematically understate crisis effects because they treat the economy's response as invariant across states.

Expanding on this idea, Caggiano, Castelnuovo, and Nodari (2017) employ a smooth-transition VAR (STVAR) framework to analyze how uncertainty shocks and monetary policy responses differ between expansions and recessions. They find that uncertainty shocks are strongly asymmetric, with mild effects in good times but severe contractionary effects in bad times. Moreover, policy responses are also state-dependent, with monetary policy being more effective at stabilizing the economy during expansions. The study highlights the importance of nonlinear, state-dependent modelling in capturing asymmetric responses to shocks. These results emphasize that both shock transmission and policy efficacy are nonlinear and depend heavily on the macroeconomic regime.

Mijatović & Schneider (2014) complement the findings above by introducing a nonlinear stochastic volatility model for the joint dynamics of the S&P 100 index and its implied volatility index (VXO). Their framework, estimated via an expected maximum likelihood algorithm, captures features such as strong mean reversion and persistent volatility more effectively than standard linear models. While their model does not improve stock return forecasts, it substantially improves volatility forecasts, particularly during extreme events, demonstrating that nonlinear specifications can better reproduce and predict observed market behaviour.

Together, these studies show that state-dependent and nonlinear models reflect financial dynamics more accurately. For asset pricing, this suggests that returns and characteristics may themselves vary and interact with macroeconomic conditions. Capturing such behaviour motivates the inclusion of interaction terms between asset traits and macroeconomic state variables in the subsequent models.

Nonlinear Factor and Characteristics Models

While state-dependent models emphasize the macroeconomic context of nonlinearities, another section of the literature focuses on how asset- or firm-level characteristics create nonlinear pricing effects. This research concerns itself with whether the cross-sectional relationships between characteristics and returns implied by standard linear factor models are consistent with empirical findings.

Kirby (2020) evaluates whether cross-sectional regression estimates of firm characteristics are consistent with the restrictions imposed by SDF models. He extends the Fama-MacBeth framework with local linear (k-nearest neighbours) regressions and finds that firm characteristics, such as size, book-to-market, profitability, capital expenditures, liquidity, and momentum, significantly predict stock returns, but in nonlinear and heterogeneous ways across firms. For example, the marginal impact of profitability or investment intensity on returns differs noticeably between low- and high-value firms. Traditional linear SDF specifications fail to reconcile these observed coefficients, implying misspecification. The study concludes that nonlinearities and omitted interaction effects are key drivers of persistent asset pricing anomalies.

Almeida and Freire (2023) further formalize the above by comparing linear and nonlinear extensions of standard factor models. Using Cressie–Read discrepancy functions, they estimate nonlinear SDFs that minimize pricing errors subject to no-arbitrage and preference constraints. Testing 19 traded factors and 44 anomaly portfolios from 1972 to 2018, they find that incorporating nonlinearities systematically raises the Sharpe ratio of the SDF’s mimicking portfolio across nearly all factor models. Strikingly, nonlinear re-estimation of simpler models such as the CAPM can enhance their performance, narrowing the gap with multifactor benchmarks. These results highlight that linear factor specifications understate risk premia because they ignore higher-order dependencies and that the relative performance of models depends strongly on how nonlinearity is specified.

Together, these studies underscore that the mapping from characteristics or factors to expected returns is inherently nonlinear. Empirically, this supports the use of transformations and interaction terms to capture curvature and conditional effects in cross-sectional return regressions. In the context of real assets, such nonlinear dependencies may arise when leverage,

duration, or cash flow stability interact to shape risk exposures, mechanisms that linear additive models cannot adequately represent.

Machine Learning Approaches

While traditional econometric models impose explicit functional forms, recent research in asset pricing has turned to machine learning (ML) methods to uncover complex, nonlinear dependencies directly from the data. These approaches replace parametric assumptions with flexible, high-dimensional estimators that can detect interactions and conditional effects among hundreds of predictors, which is a highly appealing feature for studying large datasets of asset characteristics and macroeconomic states that jointly influence returns.

Gu, Kelly, and Xiu (2020) provide one of the first systematic approaches of machine learning to asset pricing. Using data on over 30,000 U.S. stocks from 1957 to 2016 and a predictor set exceeding 900 firm characteristics and macro variables, they test a wide range of linear, penalized, and nonlinear models, including penalized regressions, tree-based models, and neural networks. Their results demonstrate that nonlinear models consistently outperform linear benchmarks in predicting returns, with complex predictor interactions driving the improvements. Among the nonlinear estimators, shallow neural networks achieve the best out-of-sample accuracy, whereas deeper models tend to overfit due to the low signal-to-noise ratio in returns. Importantly, while these ML models significantly improve predictability, they sacrifice interpretability, highlighting a trade-off between flexibility, predictability, and economic interpretation.

Chen, Pelger, and Zhu (2024) extend this approach by incorporating machine learning into a no-arbitrage framework. They estimate a nonlinear SDF using deep neural networks that combine three architectures: a feedforward network for the SDF, a recurrent LSTM to extract macroeconomic states, and a generative adversarial network (GAN) to select the most informative test assets. This design enforces theoretical constraints while flexibly capturing nonlinear interactions between firm characteristics and macroeconomic conditions. Empirically, the GAN-based SDF achieves a Sharpe ratio roughly 20% higher than benchmarks, explains over 90% of cross-sectional return variation on anomaly-sorted decile portfolios, and outperforms both linear models and prediction-based ML approaches. The results suggest that while

individual predictors impact returns nearly linearly in isolation, their interactions and dependence on macro regimes are highly nonlinear.

These two papers illustrate how flexible learning algorithms can reveal complex dependencies that traditional models overlook. However, while such deep learning frameworks constitute powerful tools for modelling high-dimensional data and its nonlinearities, their implementation is beyond the scope of this project, which focuses on transparent, regression-based methods. The relevance of this literature therefore rests on its conceptual insight: return predictability improves when models include interactions and nonlinear transformations among observable traits.

Behavioural Dynamics and Theoretical Perspectives

A complementary line of research explores the origins of nonlinear pricing dynamics through behavioural mechanisms and endogenous market feedback, rather than external shocks or structural shifts. These frameworks show that, even in the absence of changing fundamentals, interactions among agents or expectations can generate nonlinear, and sometimes chaotic, price behaviour.

Schmitt, Tramontana, and Westerhoff (2020) use a nonlinear Brock-Hommes-type model to study how interactions between fundamentalist and momentum traders create market fluctuations. Their simulations demonstrate that nonlinear switching between trading strategies creates self-reinforcing feedback, boom-bust cycles, excess volatility, and chaotic price dynamics. The authors further evaluate stabilization policies and find that interventions targeting long-run fundamentals are more effective in restoring stability than short-term countercyclical measures that reduce volatility but risk entrenching mispricing. The key result is that nonlinear dynamics can arise endogenously from trader interactions without external shocks, a finding that links differences in behaviour to persistent deviations from equilibrium pricing.

In a more structural context, Hirano and Toda (2024) develop a theoretical framework for rational asset price bubbles attached to productive real assets such as housing, land, and equities. Unlike “pure” bubbles in nonproductive assets, these bubbles can be nonstationary and evolve with economic growth and can exhibit both stabilizing and destabilizing effects. Depending on liquidity constraints and investor expectations, bubbles may crowd out productive investment by diverting savings, or conversely, crowd in innovation by financing new ventures.

The framework implies that bubble dynamics are inherently nonlinear, arising from interactions between asset fundamentals, investor expectations, and financing conditions, and that traditional pricing models systematically understate these effects.

Together, these behavioural and theoretical contributions emphasize that nonlinearities are not merely statistical noise but rather intrinsic features of market behaviour. Self-reinforcing market feedback, heterogeneous expectations, and growth-linked valuation mechanisms all contribute to complex price dynamics that contrast with linear representations. For real assets, these insights suggest that feedback between leverage, duration, and cash flow stability may produce nonlinear responses that amplify returns. Capturing such mechanisms empirically motivates the inclusion of interaction and nonlinear terms in the regression framework employed in this project.

Synthesis and Research Gap

Across all themes of the literature, a central point emerges: linear models systematically understate the complexity of asset pricing dynamics. Studies on nonlinear SDFs (Almeida et al., 2022; Almeida & Garcia, 2017; Dittmar, 2002) show that incorporating higher-order moments and transformations materially improves explanatory power relative to linear benchmarks. Research on state-dependence and regime switching (Caggiano et al., 2017; Mijatović & Schneider, 2014; Silvestrini & Zaghini, 2015) show that the same economic shock can produce vastly different effects depending on the macroeconomic regime, implying that sensitivity to risk factors is conditional. Similarly, work on nonlinear factor and firm characteristics models (Almeida & Freire, 2023; Kirby, 2020) reinforces that the relationship between firm characteristics and returns is often nonlinear, as opposed to the additive assumptions of classical models. More recent machine learning literature (Chen et al., 2024; Gu et al., 2020) confirm the findings above using flexible estimators, and behavioural and theoretical perspectives (Hirano & Toda, 2024; Schmitt et al., 2020) help explain why such nonlinearities can arise endogenously.

Despite this breadth, many of these contributions focus on equities and macro variables. Little is known about how similar, or differing, nonlinearities may arise in real assets, whose pricing mechanisms differ fundamentally from financial claims. Real assets, such as real estate and commodities, are characterized by features such as direct links to underlying physical assets, heterogeneous leverage structures, and cash flows sensitive to both operating conditions and

macroeconomic regimes. These features indicate that return dynamics could exhibit interaction effects and nonlinear dependencies among asset traits, a hypothesis that has received limited empirical testing.

3. Data & Methodology

This section describes the data sources, variable construction, and empirical modelling framework used to examine how nonlinear interactions among real asset characteristics influence return predictability. The analysis combines REIT and commodity data to capture differences in reporting structure and market frequency across real assets. Financial statement variables for REITs are collected at the quarterly level, the most consistent reporting interval across firms, while commodity data are obtained at a monthly frequency, which is sufficiently granular to capture market dynamics without introducing excessive short-term noise. The following subsections detail the preparation of each dataset, the construction of macroeconomic state variables, and the econometric models used to test for nonlinear and state-dependent relationships.

3.1 REIT data and preparation

Firm-level financial statement data for U.S. Real Estate Investment Trusts (REITs) were obtained from the Compustat North America database on WRDS (2025). REITs were identified using the Standard Industrial Classification (SIC) code 6798, and each firm's Global Industry Classification Standard (GICS) code was used to assign it to one of eight subsectors (Table 3.1). To ensure balanced coverage and data quality, the sample is restricted to 2007Q1–2024Q4, a period offering consistent financial disclosure and cross-sector representation.

Table 3.1. GICS Subsectors for Equity REITs and Descriptions

Subsector	Description
Diversified	Operations spanning two or more property types
Healthcare	Properties serving the healthcare industry (e.g., hospitals, nursing homes, and assisted living properties)
Hospitality	Hotel and resort properties
Industrial	Industrial properties (e.g., industrial warehouses, distribution properties)
Office	Office buildings and business parks
Residential	Single-family residential housing, including manufactured homes, as well as multi-family housing (e.g., apartments, student housing)
Retail	Shopping malls, outlet malls, and neighbourhood and community shopping centres
Specialized	Properties not classified elsewhere (e.g., data centres, self-storage, and other niche assets)

Note. Subsectors refer to equity REITs (GICS sub-industries). Descriptions adapted from MSCI and S&P Global (2025), Global Industry Classification Standard (GICS) Structure.

Quarterly firm-level data were cleaned and standardized before aggregation. Missing values for common shares outstanding were forward- and backward-filled, variables with excessive missingness were removed, and key variables were winsorized at the 1st and 99th percentiles to control for outliers. To reflect economic size and relevance, firm-level data were aggregated to the subsector level using market-capitalization weights computed from firms' quarterly equity values, with sums applied where appropriate. From these aggregates, six core financial ratios were constructed to capture key dimensions of REIT performance. As detailed in Table 3.2, these ratios provide a consistent panel capturing efficiency, liquidity, leverage, payout policy, profitability, and valuation dimensions of REIT performance. The resulting dataset forms the basis for the REIT baseline and interaction models described in Section 3.4.

Table 3.2. REIT Financial Ratios: Definitions

Financial Ratio	Formula	Compustat Variables	Proxy For
Asset turnover	Revenue / Total assets	saleq / atq	Efficiency
Cash-to-assets	Cash and short-term investments / Total assets	cheq / atq	Liquidity
Debt-to-assets	Total long-term debt / Total assets	dlttq / atq	Leverage
Dividend yield	Dividends per share / Share price	dvpspq / prccq	Payout Policy
FFO margin	Funds from operations / Revenue	ffoq / saleq	Profitability
FFO yield	Funds from operations / Total market value	ffoq / mkvaltq	Valuation

Note. Ratios are author-constructed in Python using quarterly data from Compustat North America via WRDS (2025). Specific formulas and Compustat variables are indicated in the middle columns.

3.2 Commodity data and preparation

Monthly commodity data were obtained from Bloomberg terminals (2025) at HEC Montréal, covering January 1999 to December 2024 for a balanced panel of eight major commodities across energy, metals, and agriculture. The monthly frequency provides sufficient granularity to capture market dynamics while limiting excessive short-term noise.

Table 3.3. Commodities Used in the Analysis by Category

Category	Commodities
Energy	Crude Oil, Natural Gas
Metals	Gold, Silver, Copper
Agriculture	Corn, Wheat, Soybeans

Note. Eight benchmark commodities across Energy, Metals, and Agriculture. Data source: Bloomberg.

For each commodity, the dataset includes futures contracts for the first six maturities (F1–F6), combined with trading volume and open interest. These series allow the construction of standardized commodity-specific traits that are comparable across commodities.

Table 3.4. Commodity Traits: Definitions

Traits	Definition / Formula	Proxy For
Open interest	Number of outstanding futures contracts	Trading activity
Trading volume	Number of contracts traded per month	Market liquidity
Front spread	$F2 - F1$	Short-term term structure slope
Front-to-back spread	$F6 - F1$ (F5 for corn and wheat)	Long-term term structure curve steepness
12-1 momentum	Sum of 11 past monthly log returns (excluding most recent month)	Return trend
12-month rolling volatility	Standard deviation of monthly log returns over past 12 months	Volatility / risk

Note. F1 and F2 denote front- and second-month futures. Corn and wheat use F5 in place of F6 due to limited coverage. Predictors are lagged one month to avoid look-ahead bias.

Monthly log returns were computed from the front-month futures contract (F1) as $\log return_t = \ln(F1_t/F1_{t-1})$, using the continuous nearest-maturity series from Bloomberg. The 12-1 momentum measure sums the prior eleven monthly log returns (excluding the most recent month), while the 12-month rolling volatility is the standard deviation monthly log returns over the preceding 12 months. Spreads are defined as differences between futures prices of different maturities, with front spread being F2 minus F1 and front-to-back spread being F6 minus F1 (F5 instead of F6 for corn and wheat due to limited F6 coverage). Open interest and trading volume were extracted directly from Bloomberg. Additional commodity-specific indicators (e.g., crack spreads, rig counts, inventories) were collected and examined but ultimately excluded to maintain a consistent trait set across all commodities.

To ensure comparability and stationarity, price levels were converted to log-differences, spreads and other typically stationary variables were retained in levels, and sparse series were linearly interpolated where necessary. Each commodity panel was aligned by identifying the first month with full data coverage and truncated to a common start date. The resulting dataset contains 312 monthly observations per commodity and forms a uniform panel suitable for the baseline and nonlinear models introduced in Section 3.4.

3.3 Macroeconomic state variables

This section describes the macroeconomic and financial indicators used to capture the aggregate macroeconomic environment influencing real asset returns. The goal is to construct a consistent, representative and interpretable macro dataset that reflects the main channels through which economy-wide conditions affect real asset returns, notably through real activity, financial and credit conditions, and uncertainty or sentiment.

All variables were obtained from publicly available sources, primarily the Federal Reserve Economic Data (FRED) database, which consolidates data series from the BEA, BLS, U.S. Census Bureau, CBOE, and Federal Reserve banks (2025). Each series was collected at quarterly and monthly frequencies to align with the corresponding REIT and commodity datasets. Sixteen series were available at quarterly frequency, while only fifteen were available at monthly frequency due to GDP growth not being obtainable at this shorter frequency.

To ensure consistency and comparability, all series were transformed to achieve stationarity or near-stationarity prior to analysis. Levels were retained for variables that are stationary and economically interpretable (e.g., spreads, standardized indices); log-differences (growth rates) were applied to trending aggregates such as GDP, CPI, and M2; and first differences were used for highly persistent interest-rate levels. These transformations mitigate spurious correlations and make coefficients in subsequent regressions more interpretable. The transformed variables were then used to perform traditional principal component analysis (PCA), which summarizes the common variation among the set of macroeconomic indicators. The resulting components serve as the observable macroeconomic state variables incorporated in the baseline and nonlinear regression models described in Section 3.4.

Real Economy Variables

The following indicators summarize the real side of the economy, capturing cyclical fluctuations in output, labour markets, and housing activity that influence real asset cash flows through demand, cost, and financing channels.

Consumer Price Index (CPI). The U.S. CPI (seasonally adjusted – SA) serves as the primary measure of inflation. The year-over-year log change is used to smooth short-term volatility and capture persistent shifts in price trends.

Gross Domestic Product (GDP). Real U.S. GDP (billions of chained 2017 USD, seasonally adjusted annual rate – SAAR) provides the broadest measure of aggregate output. Because the series is non-stationary in levels, log growth rates are retained.

Industrial Production (IP). The U.S. industrial Production Index (total output, SA) offers a complementary, more cyclical measure of real activity than GDP. Year-over-year log growth is used to represent cyclical changes in production.

Unemployment Rate. The U.S. unemployment rate (SA) captures shifts in labour market conditions. First differences are taken to remove persistence in levels and isolate cyclical variation.

Housing Indicators. U.S. Housing Starts (thousands of units, SAAR) serve as a forward-looking indicator of construction activity and are approximately stationary in levels. The S&P Case-Shiller Home Price Index (not seasonally adjusted – NSA) captures household wealth and valuation cycles; its log growth rate is retained to account for trending behaviour.

Financial Conditions and Credit Supply

These indicators capture movements in credit markets and monetary conditions that influence real asset financing costs, leverage decisions, and investor risk appetite.

Bank Credit. Total bank credit at U.S. commercial banks (billions of USD, SA) serves as a broad measure of aggregate credit supply. Because of its strong upward trend, log growth is retained.

Commercial and Industrial (C&I) Loans. C&I loans (billions of USD, SA) track bank lending to businesses and provide a more cyclical view of credit supply. Log growth is used for consistency with total bank credit.

M2 Money Supply. The M2 monetary aggregate (billions of USD, SA) measures broad liquidity in the financial system. Given its nonstationary level, log growth is used to emphasize changes in monetary conditions.

Federal Funds Rate (FFR). The effective federal funds rate (average) represents the stance of short-term monetary policy. Because levels are highly persistent, first differences are taken to reflect policy-driven rate changes.

10-Year Treasury Yield. The constant-maturity 10-year U.S. Treasury yield captures long-term borrowing costs and discount rates relevant to real asset valuation. First differences are used to represent changes in rates rather than rate levels.

Baa–Aaa Corporate Bond Spread. The spread between Moody’s Baa- and Aaa-rated corporate bond yields measures credit risk premia. The level series is stationary and retained without transformation.

Sentiment, Uncertainty, and Stress Indicators

These variables capture the psychological and volatility-related dimensions of market conditions that influence investor behaviour, household expectations, and perceived risk.

Consumer Sentiment Index. The University of Michigan Consumer Sentiment Index reflects household confidence and expectations about future economic conditions. The level series is stationary and retained without transformation.

VIX Index. The CBOE Volatility Index (VIX) measures implied volatility on S&P 500 options and serves as a forward-looking measure of market uncertainty. The level form is stationary and retained.

Chicago Fed National Financial Conditions Index (NFCI). The NFCI summarizes over one hundred financial market indicators across credit, leverage, and risk categories. The index is standardized to a zero mean, and the level series is retained despite minor structural breaks during crisis periods.

St. Louis Fed Financial Stress Index (FSI). The FSI, derived from eighteen financial market indicators, provides an alternative composite measure of systemic financial stress. The level series is stationary and is retained.

Principal Component Analysis of Macro Factors

To summarize the common variation among macroeconomic variables and avoid multicollinearity in the regression framework, principal component analysis (PCA) was applied as a dimensionality reduction technique to the transformed macro dataset series at both quarterly and monthly frequencies. The quarterly sample (2007Q1–2024Q4) aligns with the REIT dataset, while the monthly sample (1999M1–2024M12) aligns with the commodities dataset. Across both

frequencies, the first three principal components (PCs) capture the bulk of common systematic variation and present consistent economic interpretations, allowing us to use them for both real asset types.

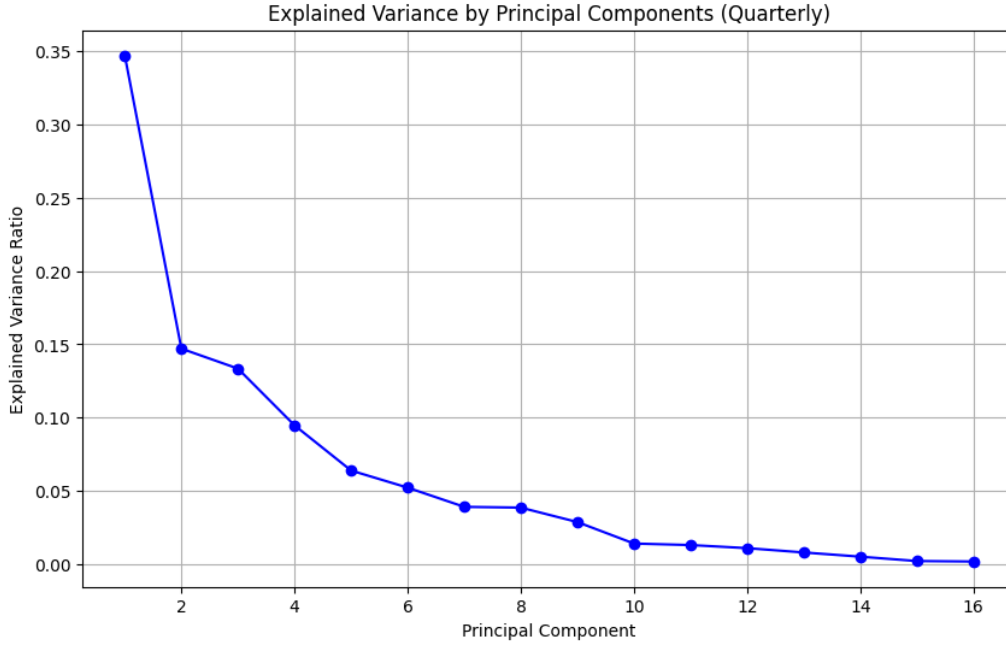
PC1 – Financial Stress and Growth ($\approx 33\%$ of variance). This principal component positively on credit spreads (Baa–Aaa), financial stress indices (NFCI, FSI), and the VIX, with negative loadings on output and housing indicators. It represents a broad “financial stress versus growth” dimension, where higher values indicate tighter financial conditions and weaker real activity.

PC2 – Credit and Liquidity Cycle ($\approx 14\%$ of variance). This PC loads positively on bank credit, business loans, M2, and housing variables, capturing fluctuations in credit expansion and liquidity associated with financial cycles.

PC3 – Prices and Uncertainty ($\approx 11\%$ of variance). PC3 loads strongly on inflation and uncertainty (VIX, spreads, policy rates), with negative contributions from consumer sentiment, reflecting a “prices and uncertainty” channel linking inflation pressures and risk perception.

Taken together, these PCs reflect three complementary dimensions of the macro-financial environment: real activity versus financial stress (PC1), liquidity and credit cycles (PC2), and inflationary or uncertainty-driven conditions (PC3). Each channel is economically relevant for real assets, which are sensitive to both financing constraints and macro demand conditions. Figure 3.1 presents the scree plot of explained variance, showing a clear inflection after the third component. The three PCs account for roughly 58% of total variance at the monthly frequency and 62% at the quarterly frequency. Higher-order PCs mainly reflect idiosyncratic or short-term noise, and interpretations differ across quarterly and monthly results. Accordingly, the first three PCs are retained as the macroeconomic state variables incorporated into the subsequent models. A full loadings heatmap illustrating the contribution of each macro variable to the first three PCs is provided in Appendix A (Figure A.1).

Figure 3.1. Scree plot of explained variance for quarterly macro principal components



Note. The plot shows explained variance ratios by principal component. There is a clear drop after PC1 and another following PC3. The monthly scree plot, showing similar results, is provided in Appendix A (Figure A.2).

3.4 Modelling framework

For both asset classes, a sequential econometric framework is employed to test for nonlinear and state-dependent dynamics in asset returns. The modelling strategy progresses from simple pooled specifications to a comprehensive framework incorporating both trait-macro interactions and nonlinear macro regime effects. All models are estimated using panel OLS with standard errors two-way clustered by entity (REIT subsector or commodity) and time, accounting for both cross-sectional and serial correlation.

Baseline Models (Models 1–3)

These models establish the baseline explanatory power of asset traits and common macroeconomic factors.

Model 1 – Traits Only (Entity Fixed Effects)

This initial specification, defined in Equation (1), regresses asset returns ($R_{i,t}$) on a vector of lagged asset traits ($Traits_{t-1}$). The coefficient vector β captures the average predictive relationship between the traits from the previous period ($t - 1$) and the return in the current

period (t). The model includes entity fixed effects (by REIT sector or by commodity), denoted with α_i , which controls for all time-invariant heterogeneity across entities (e.g., structural differences between oil and gold). As such, Model 1 estimates the β coefficients using only within-entity variation over time. All remaining, unobserved factors are captured in the error term $\varepsilon_{i,t}$.

$$R_{i,t} = \alpha_i + Traits'_{i,t-1}\beta + \varepsilon_{i,t} \quad (1)$$

Model 2 – Two-Way Fixed Effects (Time and Entity)

The second specification, shown in Equation (2), extends Model 1 by adding time fixed effects (γ_t), creating a two-way fixed effects panel model. The new γ_t term absorbs all common time-varying shocks in period t that affect all entities simultaneously (e.g., global market downturns). By controlling for both entity FE and time FE, Model 2 serves as a reference point for the explanatory power achieved when unobserved common factors are controlled for.

$$R_{i,t} = \alpha_i + \gamma_t + Traits'_{i,t-1}\beta + \varepsilon_{i,t} \quad (2)$$

Model 3 – Macro Factor Model

The third model, specified in Equation (3), replaces the time fixed effects (γ_t) from Model 2 with a vector of observable lagged macroeconomic factors. This vector, F_{t-1} contains the values of the three lagged principal components (PC1–3) described in Section 3.3 from period $t - 1$. This substitution is a critical step as it attributes the common time variation to identifiable macroeconomic drivers, making the macro state observable and enabling explicit analysis of nonlinear and state-dependent effects in subsequent models.

$$R_{i,t} = \alpha_i + F'_{t-1}\delta + Traits'_{i,t-1}\beta + \varepsilon_{i,t} \quad (3)$$

State-Dependency and Interaction Models (Models 4–6)

These models test whether the relationship between asset returns and traits varies across macroeconomic conditions and whether nonlinearities in the macro-return relationship improve explanatory power.

Model 4 – Regime-Dummy Model (Additive Effects)

This specification, defined in Equation (4), augments the Macro Factor Model (Model 3) by introducing two regime dummy variables (D_{t-1}^G and D_{t-1}^B) representing “good” and “bad” macroeconomic states, respectively. These states are defined as the upper (top 25%) and lower (bottom 25%) quartiles of the lagged PC1 distribution. The “normal” regime, representing the middle 50% of the PC1 distribution, is omitted as the reference category. The coefficients θ_G and θ_B measure the average premium or discount on asset returns during these “good” and “bad” regimes relative to the “normal” state. This model tests whether average returns differ significantly during extreme macroeconomic conditions, capturing a simple, intercept-based form of state dependency.

$$R_{i,t} = \alpha_i + F'_{t-1}\delta + Traits'_{i,t-1}\beta + \theta_G D_{t-1}^G + \theta_B D_{t-1}^B + \varepsilon_{i,t} \quad (4)$$

Model 5 – Trait-Macro Interaction Model

This specification, following an alternative route to that of Model 4, builds on the Macro Factor Model (Model 3) to test whether the predictive power of asset traits is state-dependent. It examines whether the sensitivity of returns to asset traits (the β coefficients) varies with the macroeconomic environment. This is accomplished by adding a new block of interaction terms, as shown in Equation (5).

$$R_{i,t} = \alpha_i + F'_{t-1}\delta + Traits'_{i,t-1}\beta + Z'_{i,t-1}\phi + \varepsilon_{i,t} \quad (5)$$

The new term, $Z'_{i,t-1}\phi$ represents this set of interactions. $Z_{i,t-1}$ is a vector of k interaction terms, created by multiplying an asset trait by a macro factor (e.g., $Momentum_{i,t-1} \times PC1_{t-1}$). ϕ is the corresponding vector of coefficients, and a statistically significant ϕ implies that the macroeconomic state significantly impacts the slope of a given asset trait. To determine the exact specification for REITs and commodities, a systematic grid search is performed over a set of theory-motivated interaction terms. Models including up to three interaction terms ($k \leq 3$) are estimated, and the model achieving the highest within- R^2 is selected.

Model 6 – Joint State-Dependent Model

The final and most comprehensive model, shown in Equation (6), unites the significant trait-macro interactions from Model 5 ($Z'_{i,t-1}\phi$) with the regime dummies from Model 4 ($\theta_G D_{t-1}^G$ and $\theta_B D_{t-1}^B$). This model also introduces a set of regime-slope interactions, $\lambda_G(PC1_{t-1}D_{t-1}^G)$ and $\lambda_B(PC1_{t-1}D_{t-1}^B)$. These new variables are created by interacting the $PC1_{t-1}$ factor directly with the “good” and “bad” regime dummies. The coefficients λ_G and λ_B capture the change in the slope of the PC1–return relationship during these extreme regimes, relative to the slope in the “normal” regime.

$$R_{i,t} = \alpha_i + F'_{t-1}\delta + Traits'_{i,t-1}\beta + Z'_{i,t-1}\phi + \theta_G D_{t-1}^G + \theta_B D_{t-1}^B + \lambda_G(PC1_{t-1}D_{t-1}^G) + \lambda_B(PC1_{t-1}D_{t-1}^B) + \varepsilon_{i,t} \quad (6)$$

4. Analysis of REITs

4.1 Exploratory data analysis

An exploratory data analysis (EDA) was conducted on the 2007Q1–2024Q4 quarterly panel to assess its statistical properties and inform the subsequent modelling framework. The analysis confirmed the data are well-suited for panel regression and revealed key patterns in return distributions, trait relationships, and preliminary nonlinearities.

4.1.1 Time-Series and Distributional Properties

To validate the panel OLS framework, all subsector return series were tested for stationarity and serial correlation. Formal tests confirm the data are well-suited for panel modelling. As shown in Table 4.1, across all subsectors, Augmented Dickey–Fuller (ADF) tests strongly reject the null hypothesis of a unit root ($p = 0.000$), while the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests fail to reject the null hypothesis of stationarity ($p = 0.10$). Furthermore, Ljung–Box Q(12) tests, also presented in Table 4.1, and inspection of ACF/PACF plots (see Appendix B, Figures B.1–B.2) show no evidence of persistent serial correlation.

Table 4.1. Stationarity and Serial Correlation Diagnostics for REIT Returns

Sector	ADF Test		KPSS Test		Ljung–Box Q(12) Test	
	Statistic	p-value	Statistic	p-value	Statistic	p-value
Diversified	-5.3708	0.0000	0.2333	0.1000	7.7756	0.8024
Healthcare	-8.4256	0.0000	0.1163	0.1000	11.4930	0.4872
Hospitality	-7.5299	0.0000	0.1031	0.1000	12.7945	0.3841
Industrial	-7.4280	0.0000	0.1275	0.1000	9.5342	0.6568
Office	-7.5414	0.0000	0.1046	0.1000	9.7518	0.6377
Residential	-7.1442	0.0000	0.1101	0.1000	11.9970	0.4459
Retail	-8.2290	0.0000	0.1296	0.1000	3.3500	0.9925
Specialized	-7.8394	0.0000	0.1131	0.1000	11.5167	0.4852

Note. Quarterly subsector returns, 2007Q1–2024Q4. ADF tests the null of a unit root; KPSS tests the null of stationarity; Ljung–Box Q(12) tests the null of no serial correlation up to 12 lags. Reported values are test statistics with their p-values.

While returns are stationary, they are not normally distributed. Distributional analysis via Q–Q plots (see Appendix B, Figure B.3) and formal tests confirmed significant non-normality. As shown in Table 4.2, the tail-sensitive Anderson–Darling test rejects the null of normality at the 5% significance level for all subsectors except Healthcare and Specialized. This fat-tailed behaviour is most pronounced in the Retail and Hospitality sectors, which also exhibit the highest volatility, and is consistent with typical financial return data.

Table 4.2. Normality Tests for REIT Returns (K–S and A–D)

Sector	K–S test		A–D test	
	Statistic	P-value	Statistic	Critical Value (5%)
Diversified	0.1415	0.1052	2.4582	0.749
Healthcare	0.0834	0.6759	0.5886	0.749
Hospitality	0.1615	0.0438	2.7756	0.749
Industrial	0.1135	0.2966	1.1841	0.749
Office	0.1334	0.1461	1.6703	0.749
Residential	0.0853	0.6486	1.0098	0.749
Retail	0.2174	0.002	6.9542	0.749
Specialized	0.0702	0.8508	0.4055	0.749

Note. Quarterly subsector returns, 2007Q1–2024Q4. Both tests evaluate the null of normality with mean and variance estimated from the sample. Kolmogorov–Smirnov (K–S) columns report the test statistic and its p-value. Anderson–Darling (A–D) columns report the test statistic and the 5% significance level critical value.

4.1.2 Descriptive Statistics and Trait Heterogeneity

Descriptive statistics (Tables 4.3–4.5) show significant cross-sector heterogeneity. Average quarterly returns range from 1.52% (Office) to 5.54% (Retail), with the Retail mean and extreme kurtosis driven by a single quarter. We verified the spike persists as an extreme tail event under winsorization (1st and 99th percentiles). Specialized exhibits the lowest volatility (8.47% std. dev.), likely due to the variety of its underlying assets.

Table 4.3. Quarterly REIT Returns by Subsector: Summary Statistics

Sector	Mean	Std. dev.	Min	Median	Max	Skewness	Kurtosis
Diversified	2.39%	16.15%	-36.75%	2.47%	98.41%	2.7980	17.4253
Healthcare	2.34%	11.50%	-33.32%	2.86%	29.18%	-0.6479	1.2898
Hospitality	1.99%	18.58%	-52.95%	0.84%	72.85%	0.4608	4.0808
Industrial	3.78%	16.95%	-55.77%	4.18%	57.74%	-0.2437	2.9657
Office	1.52%	13.67%	-36.07%	1.96%	37.81%	-0.2634	1.6005
Residential	2.53%	11.14%	-35.82%	3.73%	36.38%	-0.6189	2.4380
Retail	5.54%	29.03%	-47.64%	2.93%	212.67%	5.1520	37.6717
Specialized	2.98%	8.47%	-26.13%	3.34%	20.86%	-0.2060	1.0413

Note. Summary statistics for the quarterly REIT returns sample spanning 2007Q1–2024Q4.

Profitability (FFO margin) is highest in Retail and Healthcare, while Hospitality shows the lowest and most volatile margins. Leverage measures based on FFO are mechanically unstable when FFO approaches zero. Accordingly, models use debt-to-assets as the leverage proxy. Together with the non-normal return shapes documented above, these patterns motivate entity fixed effects and the use of stable, non-collinear trait proxies (see Tables 4.3–4.5; Appendix B for plots).

Table 4.4. FFO Margin by Subsector: Summary Statistics

Sector	Mean	Std. dev.	Min	Median	Max	Skewness	Kurtosis
Diversified	27.75%	15.39%	-6.87%	27.35%	51.43%	-0.1301	-1.3301
Healthcare	41.92%	6.74%	26.35%	42.91%	54.82%	-0.1942	-0.7619
Hospitality	12.34%	17.85%	-105.22%	16.38%	23.82%	-4.8069	27.7357
Industrial	41.50%	14.78%	-3.29%	47.36%	64.50%	-1.2540	1.1566
Office	37.70%	4.82%	16.05%	38.46%	46.16%	-1.7605	5.9486
Residential	35.96%	5.60%	15.64%	37.70%	44.39%	-0.9109	1.1526
Retail	48.34%	8.28%	25.08%	50.50%	58.74%	-0.9884	0.3350
Specialized	25.11%	8.21%	11.45%	29.49%	34.83%	-0.4347	-1.4963

Note. FFO margin defined as funds from operations divided by revenue, in percentage terms. The sample spans quarterly observations for 2007Q1–2024Q4. Statistics are computed from the aggregated quarter values for each subsector.

Table 4.5. Debt-to-FFO by Subsector: Summary Statistics

Sector	Mean	Std. dev.	Min	Median	Max	Skewness	Kurtosis
Diversified	11.7935	12.2014	-35.8070	7.4398	47.1515	0.2746	3.3975
Healthcare	5.2026	1.0109	3.5123	5.0563	7.8053	0.6698	0.1064
Hospitality	4.2664	38.1579	-236.3525	3.3170	208.3113	-1.4312	34.2087
Industrial	7.6501	67.1787	-429.5117	8.2029	347.9266	-2.1636	35.0556
Office	4.5977	1.8702	2.8050	3.7474	15.0362	2.8828	13.1226
Residential	3.7360	1.5829	2.0874	3.0203	10.8611	1.6784	4.3394
Retail	4.1585	1.2141	2.4536	3.6620	7.1922	0.9015	-0.2845
Specialized	4.0277	1.0572	2.5196	3.6496	6.3630	0.5473	-0.8082

Note. Debt-to-FFO defined as total debt divided by funds from operations. The sample spans quarterly observations for 2007Q1–2024Q4. Statistics are computed from the aggregated quarter values for each subsector.

4.1.3 Preliminary Nonlinearity Evidence

A primary goal of the EDA was to screen for preliminary evidence of nonlinear relationships, which are central to this project's hypothesis. Binned scatter plots and formal RESET and quadratic tests (see Appendix B, Table B.1) revealed several key patterns.

A consistent, albeit not always clear, inverted-U pattern was observed for FFO yield, with returns declining after FFO yields rise above a sector-dependent threshold, typically around 2%. Formal tests confirmed this nonlinearity was statistically significant in several sectors, notably Hospitality, Industrial, and Office. Similar nonlinearities were found for other traits but were found to be sector-dependent and thus not utilized for the purposes of this project. For example, the relationship between dividend yield and returns was nonlinear in specific sectors such as Hospitality, where very high yields appear to act as “yield traps” predictive of lower returns. Evidence was also found for sector-dependent leverage thresholds, such as in the Diversified sector where a debt-to-equity ratio exceeding 2.2x was associated with weaker subsequent returns.

4.1.4 Modelling Implications

In summary, the exploratory analysis leads to several key conclusions that help prepare the modelling strategy. First, the finding of stationary, non-autocorrelated returns validates the use of a panel OLS framework, while the presence of fat tails necessitates robust, two-way clustered standard errors. Second, the significant heterogeneity across sectors justifies the use of entity fixed effects. Finally, the evidence of nonlinearity, particularly the inverted-U pattern in FFO yield, provides a strong justification for the project's main hypothesis and the inclusion of nonlinear terms in the subsequent modelling progression.

4.2 Baseline models (R1–R3)

We begin by testing the central hypothesis that simple linear models are insufficient for explaining real asset returns. The baseline linear model, R1 (shown in Table 4.6), regresses REIT returns on the six fundamental traits—asset turnover, cash-to-assets, debt-to-assets, dividend yield, FFO margin, and FFO yield—identified in the EDA. As expected, this specification yields a low within- R^2 of 0.039, confirming that traits alone, when assumed linear, explain only a small amount of the variation in returns. While some traits like cash-to-assets ($\beta = 1.62$, $p < 0.001$) are significant, and the trait block is jointly significant (Wald $F = 4.24$, $p < 0.001$), the overall model fit is weak.

Diagnostics from the EDA provided a crucial clue, identifying a non-monotonic, “yield trap” pattern from the curvature in FFO yield. To test this, we introduce nonlinear transformations, beginning with a quadratic FFO yield term. This initial quadratic FFO yield proved to be problematic: while model fit improved modestly over the linear specification (within- $R^2 = 0.0455$) and the squared FFO yield coefficient approaches significance more closely than its linear counterpart ($p = 0.611$), its economic interpretation was flawed. The positive β_2 coefficient implies a simple U-shaped (convex) relationship, suggesting that both low and high FFO yields predict higher returns, an interpretation that directly contradicts the “yield trap” pattern identified in the EDA.

To better capture the yield trap pattern, a cubic FFO yield term was introduced. This addition not only corrects the model’s functional form to the expected inverted-S curve consistent with the yield trap, but it also improves the model’s overall joint significance (robust

F=2.37, p=0.016) and marginally increases the fit (within- $R^2 = 0.0468$). Although the individual coefficients for the squared ($\beta_2 = 84.2$) and cubic ($\beta_3 = -856.1$) terms are not yet statistically significant in this specification, the negative sign on the cubic term is essential in capturing the correct FFO yield shape. This lack of individual significance suggests the true relationship is real but is being obscured by common, time-varying market shocks.

Table 4.6. Baseline REIT Return Models (R1–R3): Panel OLS Estimates

	R1 (base)	R1 (squared)	R1 (cubed)	R2	R3
Dep. Variable	log return	log return	log return	log return	log return
Asset Turnover	-0.1457** (0.0636)	-0.0956 (0.0794)	-0.0924 (0.0714)	-0.0215 (0.0400)	-0.2054*** (0.0670)
Cash-to-Assets	1.6182*** (0.3917)	1.6249*** (0.4175)	1.6101*** (0.3961)	0.3352*** (0.1013)	1.3785*** (0.2792)
Debt-to-Assets	-0.3817* (0.1969)	-0.3374** (0.1502)	-0.3324** (0.1490)	-0.0917*** (0.0321)	-0.3118** (0.1251)
Dividend Yield	-0.4969 (2.4467)	-0.8441 (2.4199)	-1.1195 (2.5892)	-1.8661* (1.1068)	0.1531 (1.3620)
FFO Margin	-0.0106 (0.1576)	0.0872 (0.1576)	0.0916 (0.1595)	0.1612** (0.0663)	0.0170 (0.0866)
FFO Yield	2.7050 (2.5256)	1.2874 (2.5300)	2.1561 (2.5839)	1.0689 (1.3814)	5.0594* (2.6605)
Squared FFO Yield		74.364 (52.779)	84.175 (51.609)	98.529*** (27.379)	87.719*** (33.071)
Cubed FFO Yield			-856.10 (953.83)	-1267.5** (583.40)	-1373.2** (625.40)
PC1					-0.0113 (0.0106)
PC2					-0.0065 (0.0071)
PC3					-0.0238** (0.0105)
No. Observations	560	560	560	560	560
R-Squared (Within)	0.0390	0.0455	0.0468	0.0287	0.1145
F-statistic	3.6886	3.7092	3.3418	2.6879	6.3611
P-value (F-stat)	0.0013	0.0006	0.0010	0.0097	0.0000

Note. Dependent variable: quarterly REIT log returns, 2007Q1–2024Q4. All regressors lagged one quarter. Entity fixed effects included in all models; R2 additionally includes time fixed effects; R3 replaces time effects with three lagged macro principal components. Coefficients are reported with two-way clustered (by entity and time) standard errors in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

To isolate the nonlinear valuation effect, we must first control for these aggregate shocks. A standard method is to add time fixed effects (Model R2), which absorb all common time-varying shocks. As shown in Table 4.6, this step is highly effective: the cubic FFO yield expansion now becomes strongly significant ($\beta_2 = 98.5, p < 0.001$; $\beta_3 = -1267.5, p = 0.03$). Economically, these coefficients imply that returns improve as FFO yield rises from low levels, peak near 1%, and deteriorate again for very high yields. Critically, the joint F-statistic for the FFO block, using Driscoll-Kraay inference, is highly significant ($F = 36.9, p < 0.001$), confirming that this nonlinear valuation channel exists, thus confirming the existence of a “valuation corridor,” or “yield trap,” where REITs are penalized for being either too over- or under-discounted, a fact that linear models would miss.

While model R2 confirms the existence of the nonlinearity, it treats the time-varying shocks as unobservable. A more transparent approach, which becomes our new baseline (R3), replaces time dummies with three observable macroeconomic principal components. This Macro Factor Model performs well, with the within- R^2 rising to 0.115, a significant improvement over the initial R1 specification. This high R^2 also indicates that the three macro factors successfully capture much of the dispersion initially absorbed by the time dummies.

This model validates our hypothesis on two fronts. First, the cubic FFO yield structure remains highly nonlinear and statistically significant ($\beta_2 = 87.7, p = 0.008$; $\beta_3 = -1373.2, p = 0.029$), confirming the “yield trap” profile. Second, the model provides an interpretable economic link, identifying PC3 (prices/uncertainty) as a significant driver ($\beta = -0.024, p = 0.024$) of REIT returns, linking higher macro uncertainty to weaker REIT returns. This model (R3) provides a robust baseline, and its stability is confirmed by leave-one-sector-out (LOSO) robustness tests, which show the FFO yield curvature is preserved in all re-estimates. The next step we investigate is how these relationships interact with other traits, and whether they differ across different economic states.

4.3 State-dependency and interaction models (R4–R6)

The baseline model (R3) confirms that macro factors are key drivers, but it assumes their influence is continuous and uniform. We next investigate whether REIT returns exhibit more complex, state-dependent behaviour.

Table 4.7. State-Dependent and Interaction Models (R4–R6): Panel OLS Estimates

	R4	R5	R6
Dep. Variable	log return	log return	log return
Asset Turnover	-0.1569*** (0.0546)	-0.2370*** (0.0658)	-0.1897*** (0.0588)
Cash-to-Assets	1.3559*** (0.2515)	1.0123*** (0.1019)	1.1306*** (0.0918)
Debt-to-Assets	-0.3670** (0.1536)	-0.3666*** (0.1154)	-0.4096*** (0.1496)
Dividend Yield	1.3669 (1.2126)	2.4368** (1.0044)	3.7321*** (1.3589)
FFO Margin	0.0866 (0.0980)	-0.0324 (0.0843)	0.0525 (0.1077)
FFO Yield	4.0807* (2.3925)	4.5998** (1.9567)	4.0077** (1.9828)
Squared FFO Yield	121.89*** (41.183)	137.53*** (37.123)	167.99*** (46.091)
Cubed FFO Yield	-979.59* (508.01)	-2330.3*** (347.78)	
PC1	-0.0296** (0.0143)	-0.0095 (0.0098)	0.0368 (0.0397)
PC2	-0.0075 (0.0065)	-0.0073 (0.0072)	-0.0064 (0.0065)
PC3	-0.0239** (0.0098)	-0.0237** (0.0110)	-0.0225** (0.0103)
Bad State (dummy)	0.1086*** (0.0387)		0.0874** (0.0379)
Good State (dummy)	-0.0353 (0.0338)		-0.0324 (0.0357)
Debt-to-Assets × PC3		-0.1285** (0.0632)	-0.1294** (0.0590)
FFO Margin × PC1		0.0726*** (0.0176)	0.0567*** (0.0095)
PC1 × Bad State			-0.0705 (0.0435)
PC1 × Good State			-0.0486 (0.0402)
No. Observations	560	560	560
R-Squared (Within)	0.1542	0.1515	0.2057
F-statistic	7.5581	7.4016	8.1519
P-value (F-stat)	0.0000	0.0000	0.0000

Note. Dependent variable: quarterly REIT log returns, 2007Q1–2024Q4. All regressors lagged one quarter. Each model includes entity fixed effects and the cubic FFO yield expansion. R4 includes regime dummies, R5 includes interaction terms, R6 combines R4 and R5 while adding regime dummy and PC1 interactions. Coefficients are reported with two-way clustered (by entity and time) standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

We first test for discrete regime shifts (model R4) by introducing dummy variables based on PC1, separating the sample into “good” (bottom quartile) and “bad” (top quartile) macro environments. This specification substantially improves the explanatory power, with the within- R^2 rising to 0.154 (from 0.114 in R3). The regime block is jointly significant (Wald $F = 7.93$, $p = 0.019$), but it reveals a critical piece of asymmetry: the “good state” effect is insignificant, while the “bad state” dummy is positive and highly significant ($\beta = +10.9$ pp, $p = 0.005$). This finding contrasts with the negative coefficient on the continuous PC1 variable, suggesting a nonlinear relationship. Interestingly, the continuous PC1 (financial stress vs. growth axis) and PC3 (prices/uncertainty) variables remain significant drivers on their own, implying that tighter conditions continue to depress REIT returns even after accounting for the discrete regime breaks. While increasing financial stress (PC1) lowers returns, REITs appear to experience a strong positive rebound during the worst macro regimes, a dynamic consistent with mean-reverting pricing under extreme distress.

While model R4 confirms that discrete states matter, it treats all firms identically within those states. A more nuanced approach, implemented through model R5, is to test whether firm-specific traits interact with the continuous macro state. We place the discrete time dummies with two continuous interaction terms, selected as the best performing pair from a 28-specification grid: *Leverage* $\times$ *PC3* (uncertainty) and *FFO Margin* $\times$ *PC1* (stress/growth). This model achieves a comparable fit (within- $R^2 = 0.151$ vs. 0.154 in R4), and the new interaction block is strongly significant (Wald $F = 17.1$, $p = 0.0002$). This indicates that the continuous interactions capture a similar amount of underlying variation as the discrete regimes but with differing sources.

The interactions reveal clear economic channels. First, the *Leverage* $\times$ *PC3* coefficient ($\beta = -0.13$, $p = 0.043$) confirms that the penalty for high leverage intensifies during periods of high uncertainty. Second, the *FFO Margin* $\times$ *PC1* interaction ($\beta = 0.073$, $p < 0.001$) shows that operational profitability is most highly rewarded during growth phases and becomes less relevant under stress. The economic magnitudes of these effects are substantial, with one-standard-deviation shocks to the interactions accounting for 1.7–2.7% of quarterly returns. This model provides a more dynamic picture, and its stability is confirmed by LOSO robustness checks, all while the foundational “yield trap” FFO structure remains highly significant.

So far, models R4 (discrete regimes) and R5 (continuous interactions) have been presented as competing explanations. Our final specification, model R6, tests whether these effects are instead complementary by creating a hybrid model that includes both R4's regimes and R5's interactions. In addition, we interact PC1 with the "good" and "bad" state dummy variables, thus allowing the slope of PC1 to vary by regime.

The result is a pronounced gain in model fit, with the within- R^2 jumping to 0.206, a substantial increase over both prior models. The regime block is jointly significant (Wald $F = 10.47$, $p = 0.033$), with the intercept shifts alone being highly significant (Wald $F = 7.64$, $p = 0.022$) while slope adjustments remain weaker (Wald $F = 3.16$, $p = 0.206$). This strongly suggests that REIT pricing is governed by both continuous, state-dependent sensitivities and discrete, "level-shifting" regime breaks, with these effects being additive, not substitutes of each other.

This hybrid model validates our dual hypothesis. The *Leverage* $\times$ *PC3* and *FFO Margin* $\times$ *PC1* interactions remain significant, as does the "bad state" intercept ($\beta = +8.7pp$, $p = 0.022$), confirming the "rebound" effect. Furthermore, the model reveals that the PC1 slope itself is state-contingent: it flips from mildly positive in normal quarters (+0.037) to negative in bad quarters (-0.034), indicating that REITs' macro-beta is pro-cyclical. Most notably, in this final specification, the cubic FFO yield term becomes statistically redundant and is excluded, as it was found to be highly collinear with the discrete regime dummies. This is a key finding that suggests that the "yield trap" effect, so prominent in our earlier models, was effectively a proxy for the discrete "bad state" macro regime. This final model (R6) successfully integrates both forms of nonlinearity, providing the best fitting and most economically interpretable framework for our analysis.

5. Analysis of Commodities

5.1 Exploratory data analysis

Data and Return Characteristics

The commodity dataset comprises a balanced monthly panel for eight commodities from January 1999 to December 2024 (312 observations per series). For each commodity, the dataset includes futures prices for the first six maturities (F1–F6), trading volume, and open interest. From these, key traits such as spreads and momentum were constructed.

The panel exhibits significant heterogeneity in its return and risk characteristics. While average monthly returns are modest (approximately 0.2–0.7%), volatility varies substantially by group. It is most pronounced in Natural Gas (~16%) and Oil (~11%) and comparatively lower in Agriculture (~8%) and Metals (~4–9%). Consistent with many financial time series, commodity returns exhibit fat tails and slight negative skewness (Wheat being the exception), with Q-Q plots (see Appendix C, Figure C.1) confirming these departures from normality. Furthermore, volatility spikes clearly align with major macroeconomic events, including the 2008-09 financial crisis, the 2014-16 oil collapse, the COVID-19 shock, and the 2022 invasion of Ukraine. Natural Gas, in particular, consistently demonstrates the most extreme risk profile, combining high volatility with large tail losses.

Table 5.1. Monthly Commodity Log Returns: Summary Statistics

Sector	Mean	Std. dev.	Min	Median	Max	Skewness	Kurtosis
Oil	0.57%	10.99%	-78.19%	1.81%	63.33%	-0.8360	11.2412
Natural gas	0.20%	15.93%	-53.81%	0.62%	48.62%	-0.1345	1.1438
Gold	0.71%	4.65%	-19.66%	0.51%	14.88%	-0.1412	1.0119
Silver	0.56%	8.58%	-32.65%	0.10%	26.70%	-0.1335	1.0195
Copper	0.57%	7.40%	-45.37%	0.36%	30.32%	-0.5727	5.3327
Corn	0.24%	8.40%	-30.84%	0.31%	27.12%	-0.2835	1.0327
Soybeans	0.20%	7.57%	-39.85%	0.22%	18.07%	-0.7793	2.6626
Wheat	0.22%	8.78%	-29.10%	0.14%	35.30%	0.1015	0.8932

Note. Summary statistics for the monthly commodity returns sample spanning 2007M1–2024M12.

As shown by the ACF/PACF plots in Appendix C (Figures C.2 and C.3), monthly returns exhibit no significant autocorrelation, consistent with near-martingale behaviour. Conversely, trait variables such as open interest show higher levels of persistence.

Term Structure and Nonlinearity

Term structure dynamics differ systematically across the panel, as illustrated by the spread plots in Appendix C (Figure C.4). Energy markets exhibit high state-dependency, with Oil alternating between contango and backwardation based on supply shocks. In contrast, Metals like Gold typically remain in contango, while Agriculture exhibits strong seasonal oscillations.

In summary, the commodity panel exhibits pronounced cross-sectional heterogeneity, fat-tailed return distributions, and strong state dependence. Spread-based traits emerge as consistent predictors, though their effects are nonlinear and vary by macro regime. These findings motivate the nonlinear interaction and regime-switching specifications implemented in the subsequent analysis.

5.2 Baseline models (C1–C3)

This section establishes a robust baseline model for commodity returns, progressing from a simple traits-only specification to a full macro-factor model. The results, summarized in Table 5.1, are discussed below.

We begin with a standard traits-only model (C1) that includes entity (commodity) fixed effects. The model's fit is modest (within- $R^2 = 0.037$), which is typical for the high-noise environment of monthly commodity returns, though the trait block is jointly significant ($F \approx 6.3$, $p < 0.01$). The key finding from this initial model is the predictive power of the term structure. The front-to-back spread is a highly significant predictor ($p < 0.01$), confirming that a steeper curve (backwardation) forecasts higher returns. This is consistent with classic storage and convenience-yield theories. We also observe a marginally significant negative momentum effect, suggesting mild short-term mean reversion. However, the model's overall low R^2 indicates that static traits alone leave most of the monthly variation unexplained, motivating the inclusion of time-varying factors.

A primary concern with model C1 is that the observed trait effects might be spurious, merely acting as proxies for unobserved, time-varying market shocks. To test this, model C2 introduces time fixed effects to absorb common aggregate time shocks. The purpose of this model is not to increase the within- R^2 , which in fact drops slightly to 0.0321, but to perform a rigorous robustness test. The results are striking: after cleaning the model of aggregate noise, the

trait signals strengthen, with the joint F-statistic for the trait block rising sharply (Wald $F \approx 32.6$, $p < 0.01$) from model C1's 6.3. The front-to-back spread remains highly significant ($p < 0.01$), the front spread becomes newly significant ($p < 0.05$), and the negative momentum effect increases in significance ($p < 0.05$). This is a crucial finding which implies that the C1 model was understating the importance of these traits by confounding them with macro noise. We can therefore conclude that these term structure and momentum signals are robust, non-spurious cross-sectional predictors and not simply artifacts of shared macro risks.

Table 5.2. Baseline Commodity Return Models (C1–C3): Panel OLS Estimates

	C1	C2	C3
Dep. Variable	log return	log return	log return
Open Interest	-0.0010 (0.0024)	-0.0010 (0.0023)	-0.0006 (0.0024)
Trading Volume	-0.0002 (0.0019)	0.0025 (0.0024)	0.0010 (0.0019)
Front Spread	0.0002 (0.0002)	0.0004** (0.0002)	0.0002 (0.0001)
Front-to-Back Spread	0.0003*** (0.0001)	0.0002*** (6.177e-05)	0.0003*** (0.0001)
12-1 Momentum	-0.0149* (0.0081)	-0.0195** (0.0083)	-0.0134 (0.0082)
12-Month Rolling Volatility	-0.0039 (0.1215)	-0.0755 (0.0967)	-0.0008 (0.0993)
PC1			6.632e-05 (0.0015)
PC2			0.0040** (0.0016)
PC3			-0.0032* (0.0018)
No. Observations	2483	2483	2475
R-Squared (Within)	0.0371	0.0321	0.0427
F-statistic	15.845	10.617	12.191
P-value (F-stat)	0.0000	0.0000	0.0000

Note. Dependent variable: monthly commodity log returns, 1999M1–2024M12. All regressors lagged one month. Entity fixed effects included in all models; C2 additionally includes time fixed effects; C3 replaces time effects with three lagged macro principal components. Coefficients are reported with two-way clustered (by entity and time) standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

While C2 confirms the signals are robust, its time dummies are a “black box.” Model C3 replaces these dummies with our three observable macroeconomic principal components (PC1–

3) to provide an interpretable economic link. This Macro Factor Model performs well, achieving the highest fit of the baseline set (within- $R^2 = 0.043$). Both the trait block (Wald $F \approx 38.0$, $p < 0.01$) and the macro PC block are highly significant (Wald $F \approx 13$, $p < 0.01$), confirming the PCs are an effective substitute for the time dummies.

This model provides a clear economic narrative: on the macro side, PC2 (credit/liquidity) is positive and significant ($p \approx 0.016$), while PC3 (price/uncertainty) is negative and marginally significant. This suggests that, as expected, high liquidity states support commodity premia while high uncertainty depresses them. On the trait side, the front-to-back spread remains the most reliable predictor ($p \approx 0.001$). Leave-one-commodity-out (LOCO) robustness tests confirm these findings are stable and not driven by any single commodity. Model C3 thus provides our robust baseline, confirming that commodity returns are driven both by their internal term structure dynamics and by their sensitivity to aggregate macro conditions.

5.3 State-dependency and interaction models (C4–C6)

This section builds on the baseline models by testing for nonlinearities and state-dependence in commodity returns. The baseline model (C3) confirmed that macro factors are drivers, but it assumed their influence is linear and constant. We now test this assumption. The models, summarized in Table 5.2, progress from a simple additive regime model (C4) to a continuous trait-interaction model (C5), and finally to a comprehensive hybrid model (C6) that integrates both discrete regimes and continuous interactions.

We first test for discrete regime shifts (Model C4) by adding “good” and “bad” state dummies based on PC1 (financial stress vs. growth axis) to the C3 baseline. Regimes are defined by PC1 quartiles (good: $PC1 \leq -1.1931$; bad: $PC1 \geq 0.566$). This specification modestly improves the fit (within- $R^2 \approx 0.0474$ vs. 0.0427 in C3), and the regime block is jointly significant (Wald $F = 8.02$, $p \approx 0.02$).

Table 5.3. State-Dependent and Interaction Models (C4–C6): Panel OLS Estimates

	C4	C5	C6
Dep. Variable	log return	log return	log return
Open Interest	-0.0007 (0.0024)	-0.0005 (0.0024)	-0.0007 (0.0024)
Trading Volume	0.0009 (0.0018)	0.0012 (0.0018)	0.0010 (0.0017)
Front Spread	0.0002* (0.0001)	0.0002* (0.0001)	0.0003** (0.0001)
Front-to-Back Spread	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)
12-1 Momentum	-0.0170** (0.0081)	-0.0063 (0.0077)	-0.0114 (0.0071)
12-Month Rolling Volatility	-0.0186 (0.0949)	-0.0234 (0.0555)	-0.0441 (0.0530)
PC1	-0.0003 (0.0020)	-0.0007 (0.0018)	0.0139** (0.0066)
PC2	0.0036** (0.0017)	0.0036** (0.0017)	0.0043** (0.0018)
PC3	-0.0036* (0.0019)	-0.0027 (0.0017)	-0.0020 (0.0019)
Bad State (dummy)	0.0127 (0.0085)		0.0130 (0.0080)
Good State (dummy)	0.0141** (0.0063)		0.0138 (0.0138)
12-1 Momentum $\times$ PC1		-0.0068* (0.0038)	-0.0081** (0.0038)
12-Month Rolling Volatility $\times$ PC3		-0.0878*** (0.0204)	-0.0796*** (0.0204)
PC1 $\times$ bad state			-0.0177** (0.0076)
PC1 $\times$ good state			-0.0094 (0.0091)
No. Observations	2475	2475	2475
R-Squared (Within)	0.0474	0.0500	0.0574
F-statistic	11.098	11.761	9.9465
P-value (F-stat)	0.0000	0.0000	0.0000

Note. Dependent variable: monthly commodity log returns, 1999M1–2024M12. All regressors lagged one month. All models include entity fixed effects, the trait block, and macro PCs. C4 includes regime dummies, C5 includes interaction terms, C6 combines C4 and C5 while adding regime dummy and PC1 interactions. Coefficients are reported with two-way clustered (by entity and time) standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The key finding is a distinct asymmetry in state-dependence: the “good state” dummy significantly lifts monthly returns by about 1.41 pp ($p = 0.026$), while the “bad state” effect is statistically insignificant ($p = 0.137$). This suggests that commodity premia rise markedly in favourable macro environments but do not have a symmetric negative reaction in stressed periods. The failure of the continuous PC1 variable to capture this on its own motivates exploring more complex interactions.

An alternative hypothesis is that state-dependence is not a discrete shift but a continuous interaction. Model C5 tests this by replacing C4’s dummies with two continuous interaction terms, identified via a grid search as the most significant pairing: *Momentum* $\times$ PC1 (stress/growth) *Volatility* $\times$ PC3 and (uncertainty).

This model achieves a superior fit (within- $R^2 = 0.050$), and the interaction block is highly significant (Wald $F = 18.76$, $p < 0.01$). The results provide clear economic channels. The most significant effect is the *Volatility* $\times$ PC3 interaction ($p < 0.001$), a strongly negative coefficient demonstrating that high past volatility is especially penalized when macro uncertainty is high, consistent with spikes in risk aversion. Concurrently, the *Momentum* $\times$ PC1 interaction ($p \approx 0.07$) suggests that momentum strategies are penalized when financial conditions improve (i.e., when PC1 is low). The economic magnitudes are substantial, with one-standard-deviation shocks to these interactions depressing returns by -0.43% and -0.50%, respectively. The stability of these interaction coefficients is confirmed by LOCO (leave-one-commodity-out) robustness tests, supporting their generalizability across the panel.

Model C4 (discrete shifts) and C5 (continuous interactions) have been presented as competing explanations. Our final specification, Model C6, tests if they are instead complementary by creating a hybrid model that includes both C4’s regime dummies and C5’s interaction terms, while also allowing the slope of PC1 to vary by regime.

This hybrid model yields a decent gain in fit (within- $R^2 = 0.0574$), suggesting that both discrete and continuous nonlinearities play a role in commodity returns. The decomposition of regime effects (Wald $F = 10.14$) reveals a crucial insight: the significance is now driven by the new PC1 slope shifts (Wald $F = 5.48$, $p = 0.065$), while the C4-style intercept shifts now become insignificant ($p = 0.210$).

This final model clarifies the true nature of state-dependence. The “good state” premium we saw in C4 was not a simple level shift, but a proxy for a more complex structural change: the

macro-beta of commodities is state-contingent. This is reinforced by the main PC1 effect, which now becomes positive and significant ($p \approx 0.04$) on its own. The PC1 slope flips from positive in “good” states (+0.0046) to negative in “bad” states (-0.0038), signalling that tightening credit conditions depress returns only during stress episodes. Furthermore, the model confirms that the uncertainty penalty is almost entirely captured by the *Volatility* $\times$ *PC3* interaction, as the PC3 main effect becomes insignificant. Overall, C6 confirms that state-dependent macro pricing in commodities is driven by a combination of: (1) robust term structure signals, (2) a pro-cyclical macro-beta (slope) that flips during stress, and (3) powerful interactions that penalize volatility during uncertainty and momentum during growth.

6. Discussion & Conclusion

This project investigated whether the linear, additive models traditionally used in asset pricing are sufficient for real assets, or if, like equities, their returns are better explained by nonlinear dynamics and state-dependence. By constructing two distinct panel datasets for REIT subsectors and major commodities, this study systematically tested the incremental explanatory power of nonlinear traits, macroeconomic regime shifts, and state-dependent interactions. The results provide a clear and consistent answer: nonlinear and state-dependent models materially outperform traditional linear frameworks in explaining real asset returns.

Key Findings

The modelling progression across both asset classes demonstrated a significant improvement in explanatory power as nonlinearities were introduced.

For REITs: The final baseline model (R3), which included traits and continuous macro factors, yielded a within- R^2 of 0.1145. The final hybrid model (R6), which incorporated both continuous interactions and discrete regime shifts, achieved a within- R^2 of 0.206, nearly doubling the explanatory power. The key findings were:

- **Nonlinear Valuation:** The relationship between FFO Yield and returns was robustly nonlinear. The data supported a cubic, inverted-S-shaped curve, consistent with a “valuation corridor.” This implies that markets penalize both “overpriced” (very low yield) and “distressed” (very high yield) REITs, with a performance peak at moderate yield levels.
- **State-Dependent Trait Premia:** The value of fundamental traits was not static. The Leverage $\times$ PC3 interaction was highly significant, showing that the negative penalty for leverage is most severe during periods of high macro uncertainty and rising rates. Conversely, the Profitability $\times$ PC1 interaction showed that strong FFO margins were most highly rewarded during “good” macroeconomic states.
- **Asymmetric Regime Effects:** REIT returns exhibit asymmetric “snap-back” behaviour. While returns fall as financial stress (PC1) builds, they show a strong positive rebound within the worst macro regimes, consistent with post-distress mean reversion.

For Commodities: A similar, though more modest, pattern emerged. The final baseline macro model (C3) yielded a within- R^2 of 0.043, which rose to 0.057 in the final hybrid model (C6).

- **Robust Trait Effects:** The Front-to-Back Spread (term structure slope) was the most consistent and significant trait, confirming that steeper backwardation predicts higher returns, in line with classic convenience yield theory.
- **State-Dependent Interactions:** The key nonlinearities were Volatility $\times$ PC3 (a strong negative) and Momentum $\times$ PC1 (a negative). This indicates that high volatility is most severely penalized during periods of high macro uncertainty, and that trend-following momentum strategies are penalized (i.e., they fail or reverse) when fundamental growth conditions are strong.
- **Macro-Beta Flipping:** The final C6 model revealed that the beta of commodities to financial conditions (PC1) is itself state-dependent, flipping from positive in good states to negative in bad states.

Discussion and Implications

The findings of this project provide strong empirical evidence that the insights from the recent equity pricing literature, most notably that nonlinearities and interactions are critical, are directly applicable to real assets. This work extends the findings of Dittmar (2002), Gu et al. (2020), and Kirby (2020) by showing that these complex dynamics are not unique to equities.

The “valuation corridor” for REIT FFO yields is a particularly noteworthy finding. It suggests that linear valuation models are misspecified. For REITs, there is an optimal valuation band. Being too “cheap” (high yield) is not a simple value signal but is correctly priced as a signal of distress, a “yield trap.”

Furthermore, the state-dependence of risk premia is a central theme. The economic intuition is clear: leverage is riskier when uncertainty is high, and profitability is a greater asset when growth is high. This supports a dynamic view of risk management, where linear, static hedge ratios or risk-exposure models would fail.

A key difference emerged between the two asset classes. The models explained a much larger portion of the variance in REIT returns (up to 20.6%) than in commodity returns (up to 5.7%). A plausible explanation, which serves as a limitation, is the U.S.-centric nature of the macroeconomic factors. The principal components were derived from U.S. macro data, which

serve as strong direct drivers for the U.S. REIT market. Commodities, however, are priced in a global market. U.S. macro conditions are only one part of their pricing story, making the factors a less powerful (though still significant) explanatory variable.

Limitations and Future Research

This analysis has several limitations that provide clear avenues for future studies:

1. **Macro Factor Specification:** As discussed, the U.S. macro factors are a limitation for commodities. Future work could construct a separate set of global macro factors (e.g., including the China PMI, the Baltic Dry Index, and European industrial production) to better capture the global drivers of commodity prices.
2. **Econometric Models:** This project used discrete regime dummies (quartiles of PC1) to capture state-dependence. A natural extension would be to employ more formal regime-switching models, such as Markov-Switching (MS-VAR) or Smooth Transition Autoregressive (STAR) to model these transitions endogenously.
3. **Factor Construction:** The macro factors were constructed using standard linear PCA. Future analysis could explore Nonlinear PCA (NLPC), as proposed by Almeida et al. (2022), to extract nonlinear factors from the macro variables.
4. **Scope of Assets:** The findings for listed U.S. REITs provide interesting hypotheses for the private real estate market. A valuable extension would be to apply this framework to private real estate data to see if these same nonlinear dynamics hold in less liquid, non-listed markets.

Concluding Remarks

This project successfully demonstrated that incorporating nonlinearities and state-dependence is critical for modelling real asset returns. The findings challenge the sufficiency of simple linear models, showing that the premia for key traits like leverage, profitability, and volatility are not static but vary significantly with the macroeconomic regime. The framework developed here, particularly the hybrid model (R6/C6) that combines continuous interactions with discrete regime shifts, proved to be a powerful tool. This research provides a strong empirical foundation for a more dynamic and nonlinear approach to valuation and risk management in the real asset space.

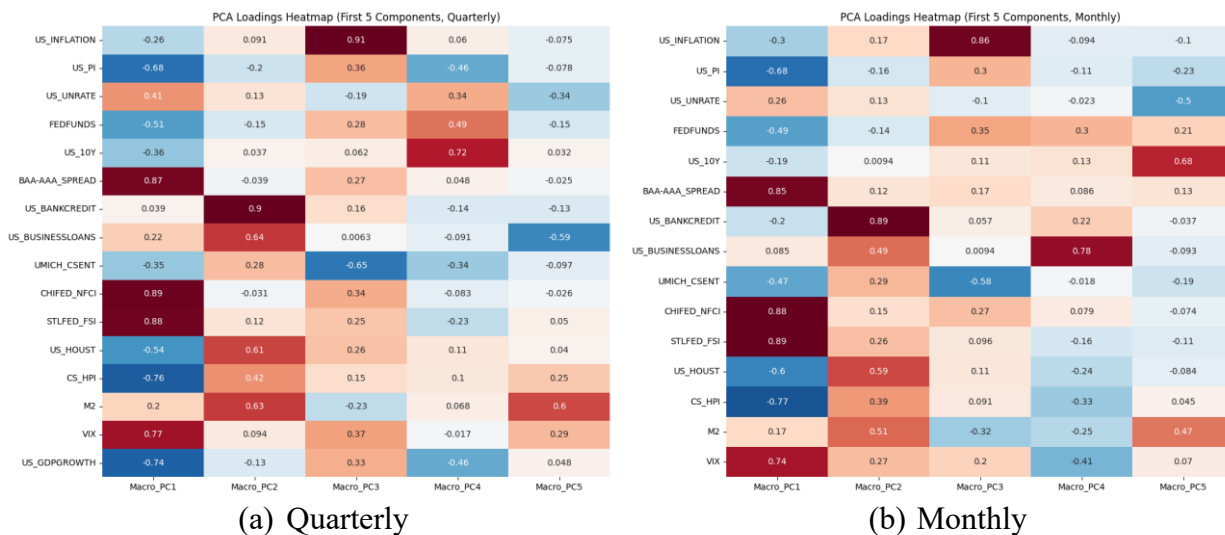
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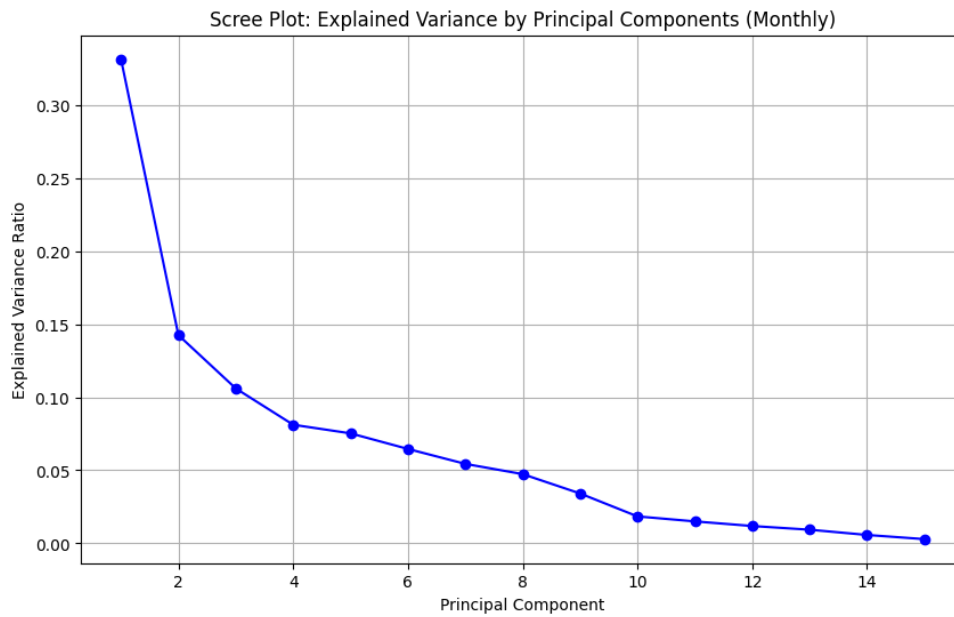
Appendix A – Macro PCA: Supplementary Figures and Tables

Figure A.1. PCA loadings for quarterly and monthly macroeconomic variables



Note. This heatmap displays the correlation loadings of the 16 quarterly and 15 monthly macroeconomic variables on the first five principal components. The first three components (Macro_PC1, Macro_PC2, and Macro_PC3) are retained as the main macroeconomic state variables due to their consistent interpretation across frequencies.

Figure A.2. Scree plot of explained variance by monthly principal component



Note. Figure A.2 shows the percentage of explained variance by monthly principal component. There is a clear drop after PC1. The drop after PC3 is less clear than for the quarterly series, but the consistency in interpretation is the key feature driving the decision to retain the first three PCs.

Appendix B – REITs: Supplementary Figures and Tables

Table B.1. Nonlinearity Diagnostics for FFO Yield: Quadratic and RESET p-values by Subsector

Subsector								
Test	Diversified	Healthcare	Hospitality	Industrial	Office	Residential	Retail	Specialized
Quadratic	0.1421	0.4040	0.0011	0.8305	0.0175	0.9209	0.9687	0.4152
RESET	0.2933	0.1849	0.0063	0.0301	0.0066	0.4002	0.9314	0.2935

Note. Quarterly REIT returns, covers 2007Q1–2024Q4. “Quadratic” reports the p-value from a robust t-test of the squared FFO yield coefficient after adding the term to a linear baseline OLS regression. “RESET” is the Ramsey misspecification test with power = 2.

Table B.2. Baseline REIT Models (R1–R3): Robust F-statistics and Poolability Tests

	R1 (base)	R1 (squared)	R1 (cubed)	R2	R3
Robust F-statistic	4.2375	1.9069	2.3739	1.2508	2.3783
p-value (robust F)	0.0004	0.0663	0.0161	0.2733	0.0070
F-test for poolability	2.5659	1.8269	1.9157	14.385	2.7984
p-value (poolability)	0.0131	0.0798	0.0650	0.0000	0.0072

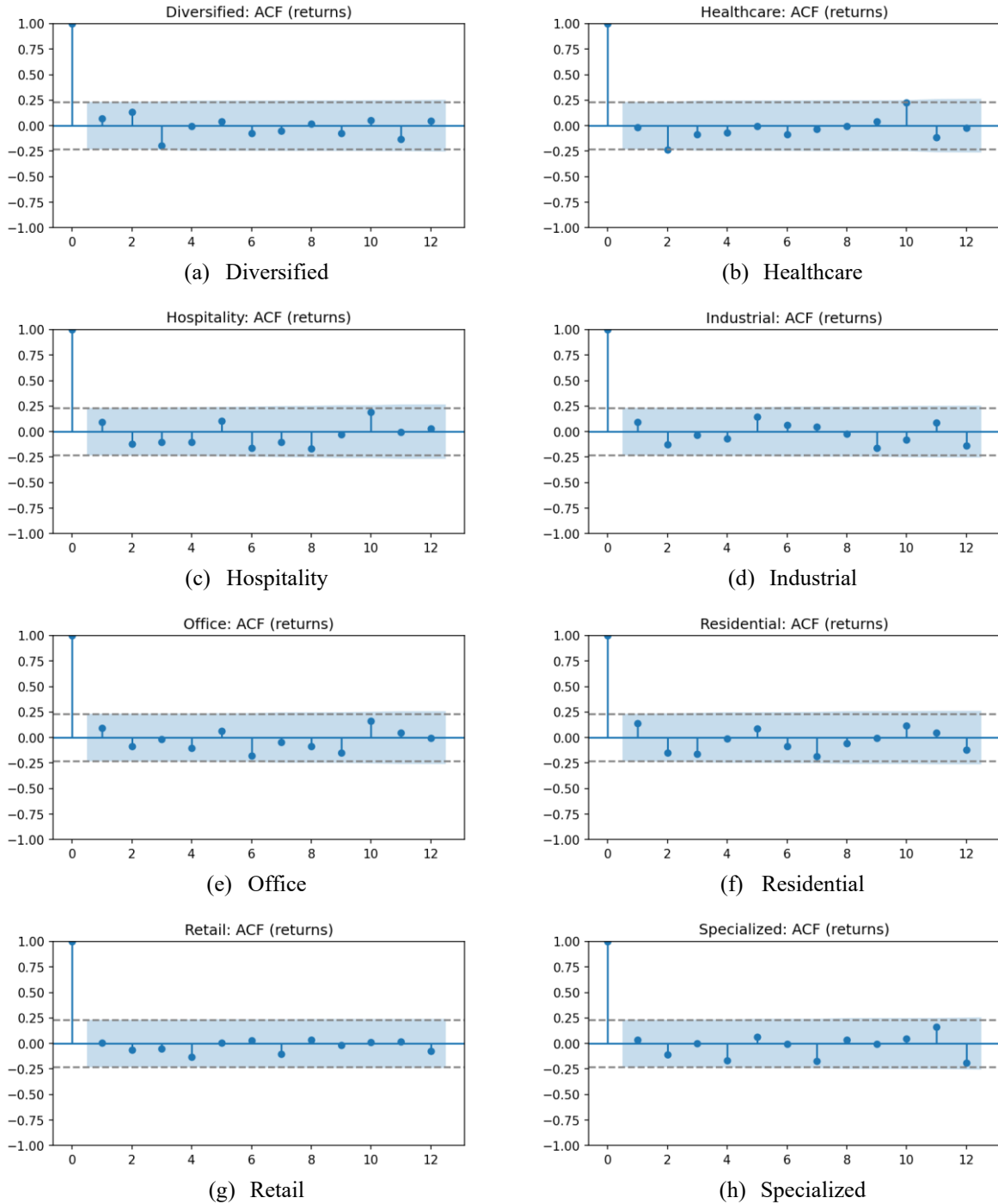
Note. Dependent variable: quarterly REIT log returns, 2007Q1–2024Q4. All regressors lagged one quarter. Entity fixed effects included in all models; R2 additionally includes time fixed effects; R3 replaces time effects with three lagged macro principal components. Coefficients are reported with two-way clustered (by entity and time) standard errors in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

Table B.3. State-Dependent REIT Models (R4–R6): Robust F-statistics and Poolability Tests

	R4	R5	R6
Robust F-statistic	10.405	3.0909	4.0340
p-value (robust F)	0.0000	0.0002	0.0000
F-test for poolability	2.8243	3.2575	3.5425
p-value (poolability)	0.0067	0.0021	0.0010

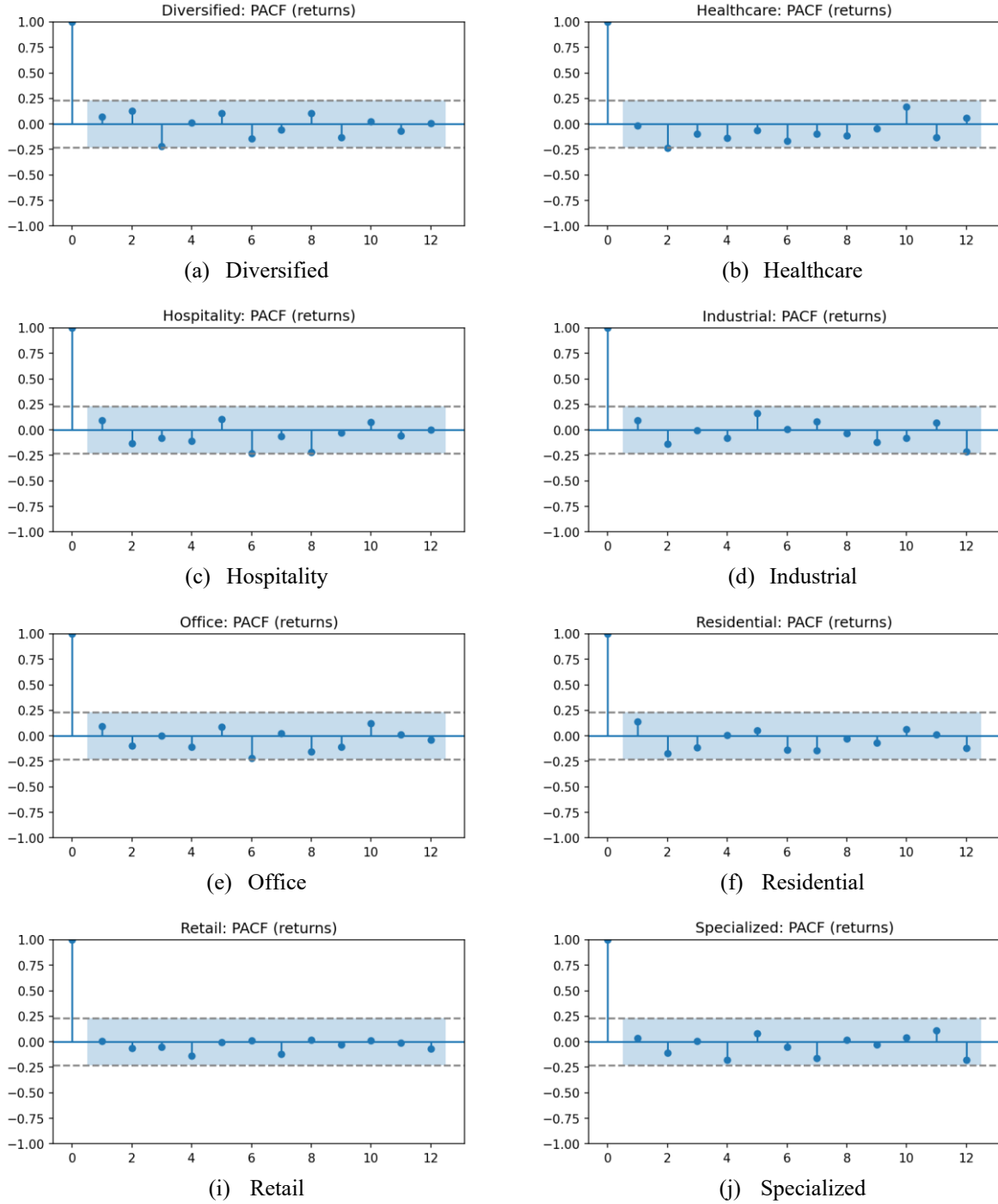
Note. Dependent variable: quarterly REIT log returns, 2007Q1–2024Q4. All regressors lagged one quarter. Each model includes entity fixed effects and the cubic FFO yields expansion. R4 includes regime dummies, R5 includes interaction terms, R6 combines R4 and R5 while adding regime dummy and PC1 interactions. Coefficients are reported with two-way clustered (by entity and time) standard errors in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

Figure B.1. ACF of quarterly REIT returns by subsector



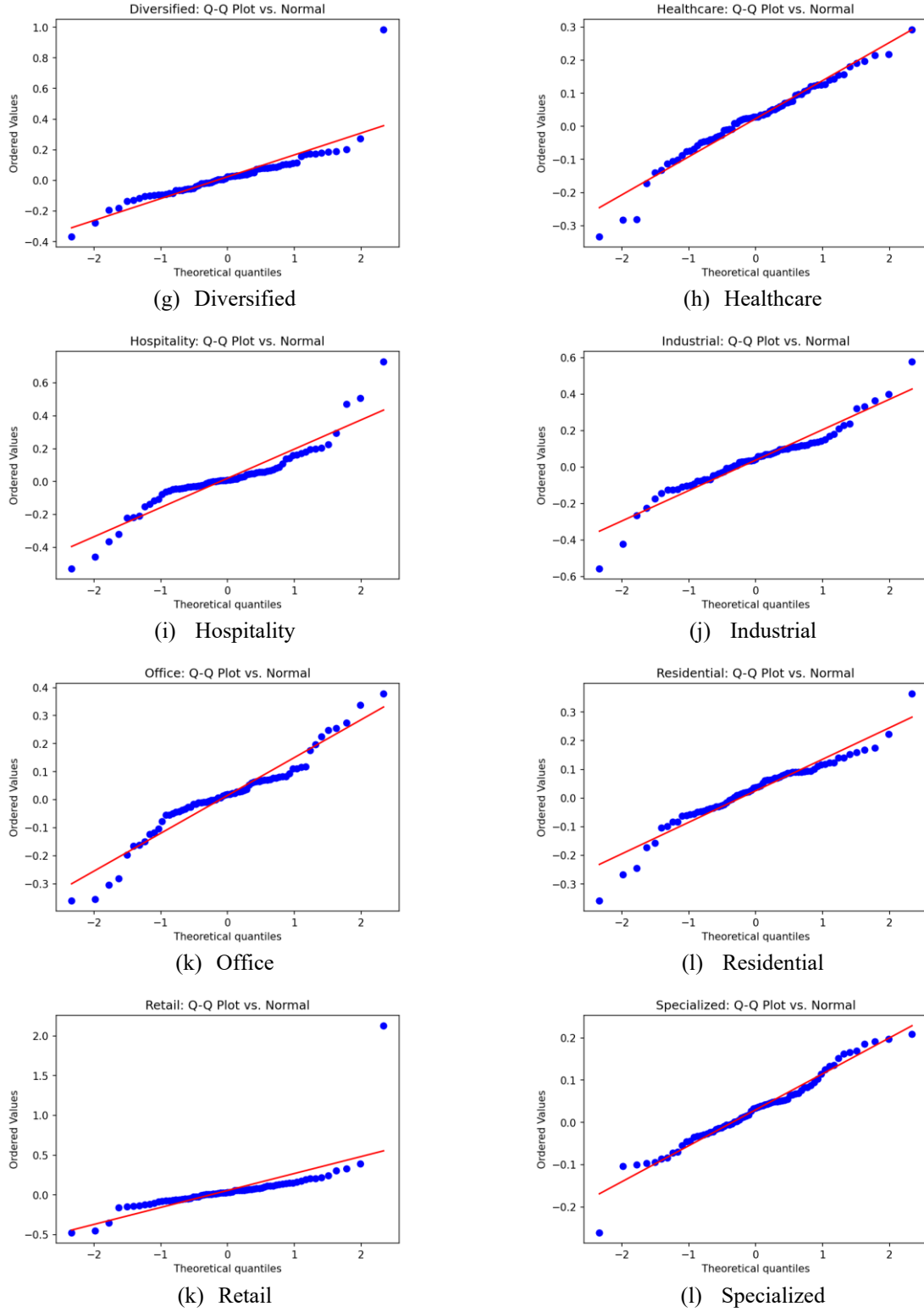
Note. Autocorrelation plots through 12 lags with 95% confidence bands. Sample spans 2007Q1–2024Q4.

Figure B.2. PACF of quarterly REIT returns by subsector



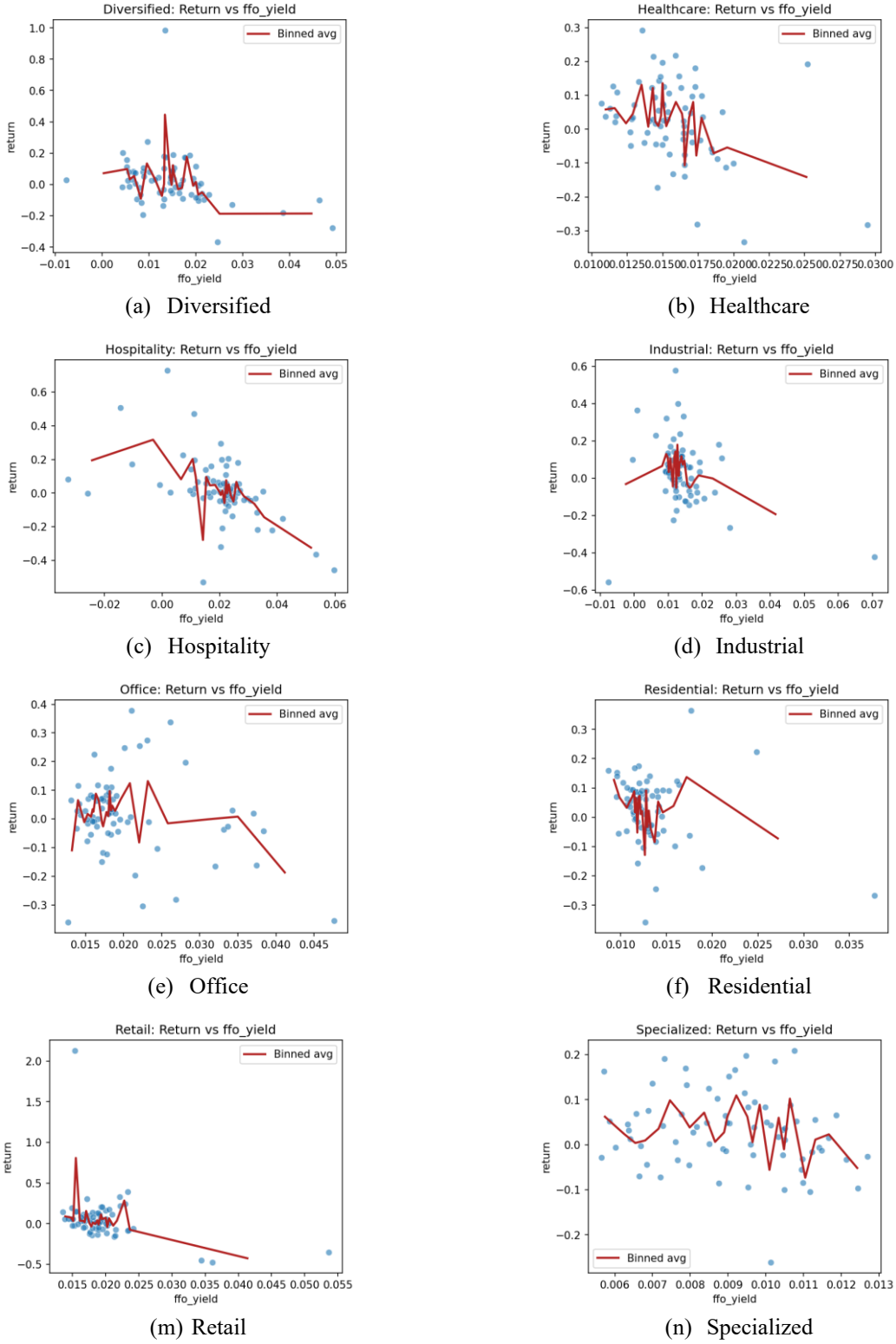
Note. Partial autocorrelation plots through 12 lags with 95% confidence bands. Sample spans 2007Q1–2024Q4.

Figure B.3. Q–Q plots of quarterly REIT returns versus normal distribution by subsector



Note. Quantiles use sample mean and variance. Deviations in the tails indicate non-normality consistent with A–D rejections in Table 4.2.

Figure B.4. Binned scatter plots of FFO yield versus REIT returns by subsector



Note. Each panel plots raw observations and a binned-average line that connects the computed bins. No smoothing is applied, the red line is a simple piecewise connection of bins.

Appendix C – Commodities: Supplementary Figures and Tables

Table C.1. Baseline Commodity Models (C1–C3): Robust F-statistics and Poolability Tests

	C1	C2	C3
Robust F-statistic	6.2798	5.4261	7.2028
p-value (robust F)	0.0000	0.0000	0.0000
F-test for poolability	1.7903	2.6780	1.7415
p-value (poolability)	0.0849	0.0000	0.0950

Note. Dependent variable: monthly commodity log returns, 1999M1–2024M12. All regressors lagged one month. Entity fixed effects included in all models; C2 additionally includes time fixed effects; C3 replaces time effects with three lagged macro principal components. Coefficients are reported with two-way clustered (by entity and time) standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.2. State-Dependent Commodity Models (C4–C6): Robust F-statistics and Poolability Tests

	C4	C5	C6
Robust F-statistic	8.7075	18.625	N/A
p-value (robust F)	0.0000	0.0000	N/A
F-test for poolability	1.7219	1.6876	1.6852
p-value (poolability)	0.0994	0.1074	0.1080

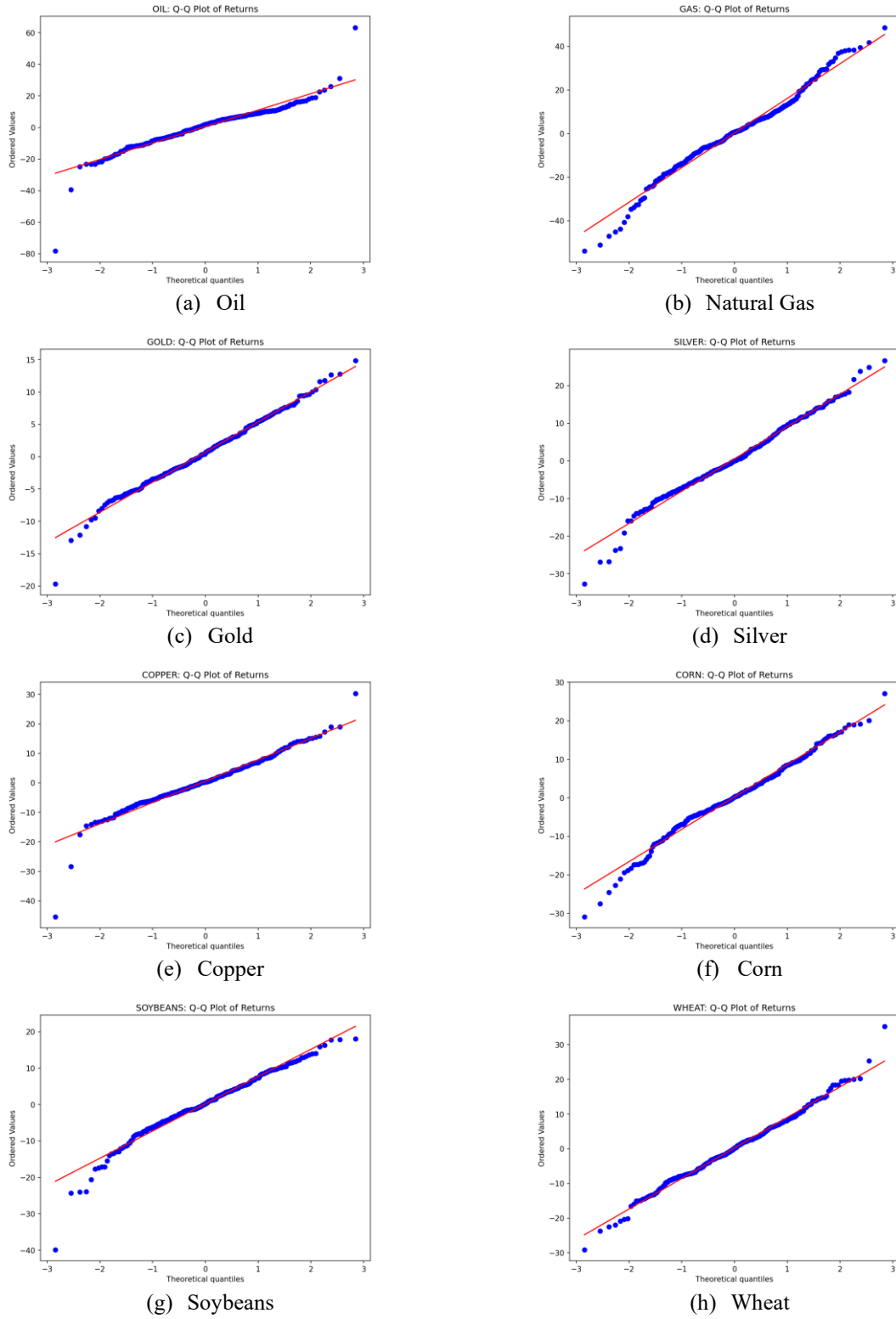
Note. Dependent variable: monthly commodity log returns, 1999M1–2024M12. All regressors lagged one month. All models include entity fixed effects, the trait block, and macro PCs. C4 includes regime dummies, C5 includes interaction terms, C6 combines C4 and C5 while adding regime dummy and PC1 interactions. Coefficients are reported with two-way clustered (by entity and time) standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.3. Block Wald Tests for Traits and Macro Factor (C1–C3)

	C1	C2	C3
Wald F-statistic (Traits block)		32.556	38.013
p-value (traits block)		0.0000	0.0000
Wald F-statistic (Macro block)			12.956
p-value (macro block)			0.0047

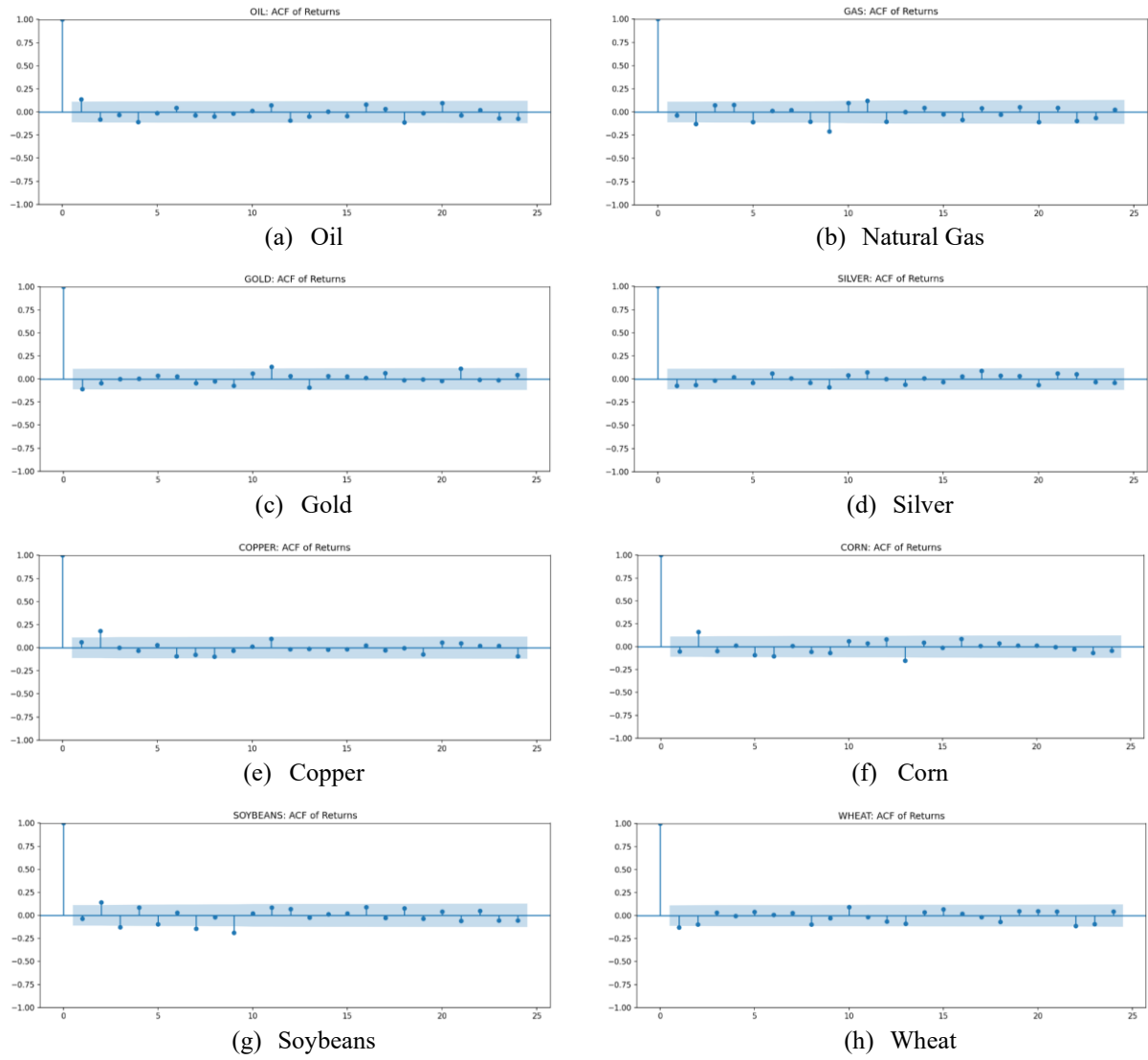
Note. Dependent variable: monthly commodity log returns, 1999M1–2024M12. Wald F-statistics and p-values for (i) the trait block and (ii) the macro block, where applicable. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure C.1. Q–Q plots of monthly commodity returns versus normal distribution



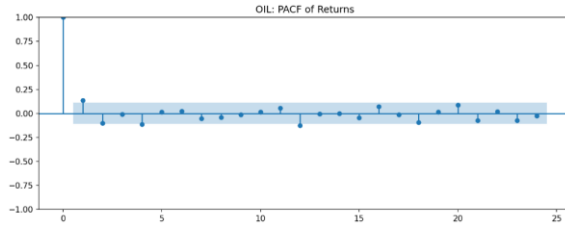
Note. Quantiles use sample mean and variance. Deviations in the tails indicate non-normality.

Figure C.2. ACF of monthly commodity returns

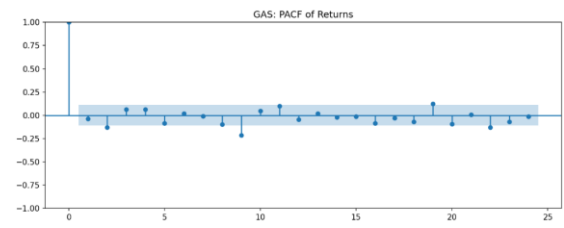


Note. Autocorrelation plots through 24 lags with 95% confidence bands. Sample spans 1999M1–2024M12.

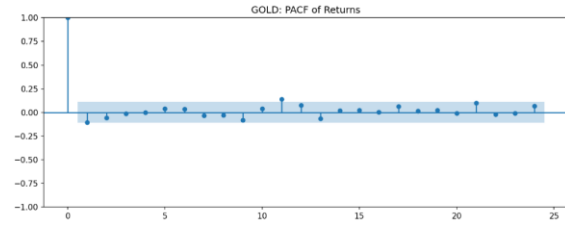
Figure C.3. PACF of monthly commodity returns



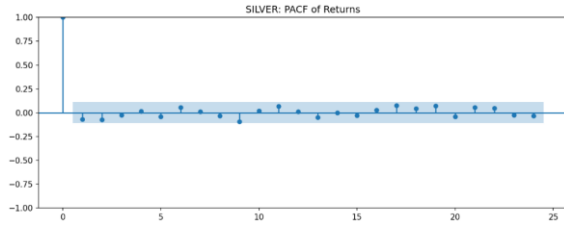
(a) Oil



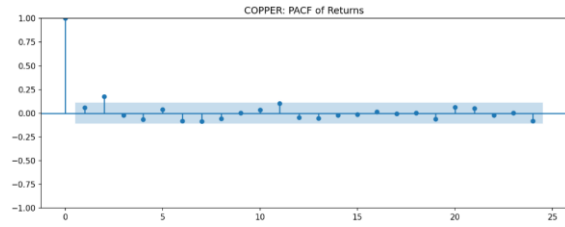
(b) Natural Gas



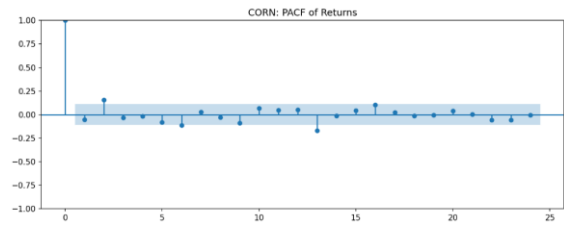
(c) Gold



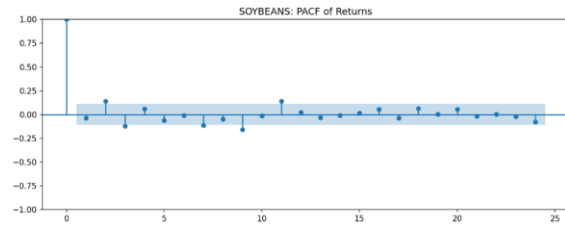
(d) Silver



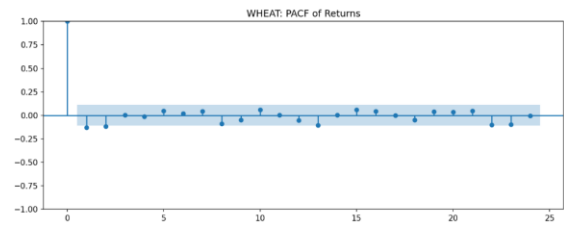
(e) Copper



(f) Corn



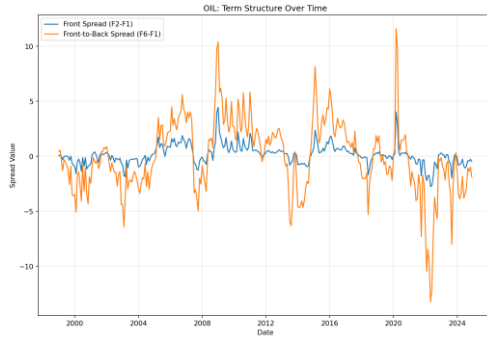
(g) Soybeans



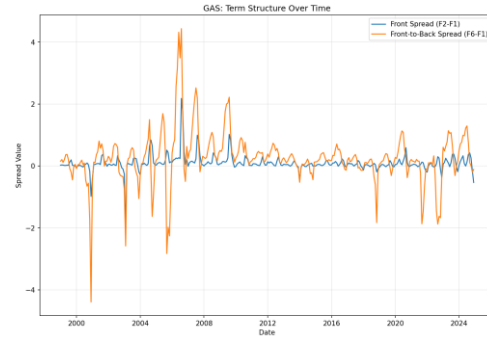
(h) Wheat

Note. Partial autocorrelation plots through 24 lags with 95% confidence bands. Sample spans 1999M1–2024M12.

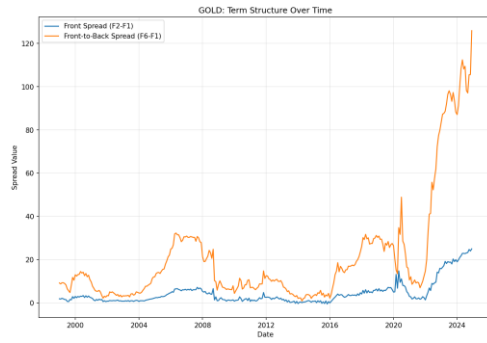
Figure C.4. Commodity term structure spreads over time (front and front-to-back)



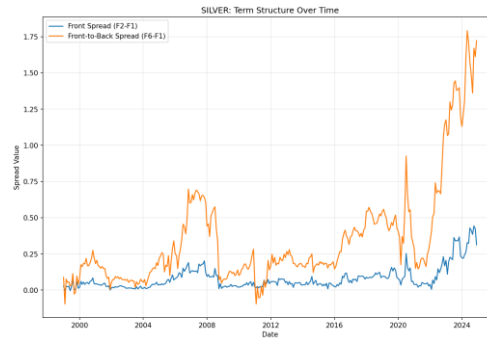
(a) Oil



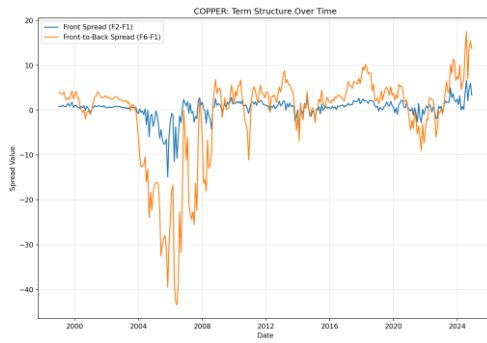
(b) Natural Gas



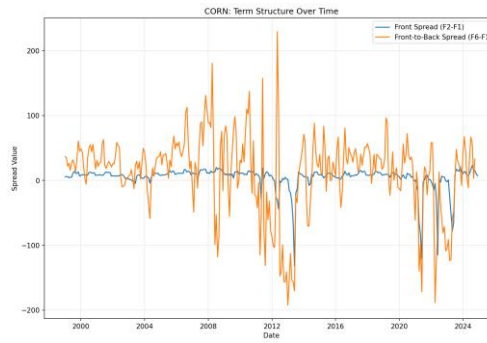
(c) Gold



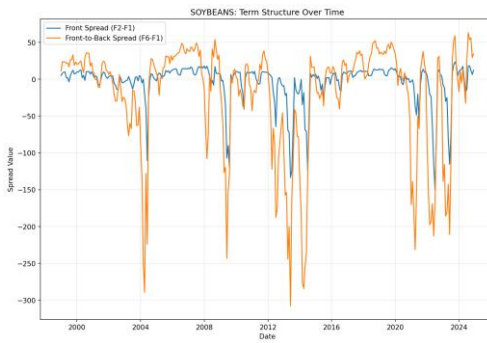
(d) Silver



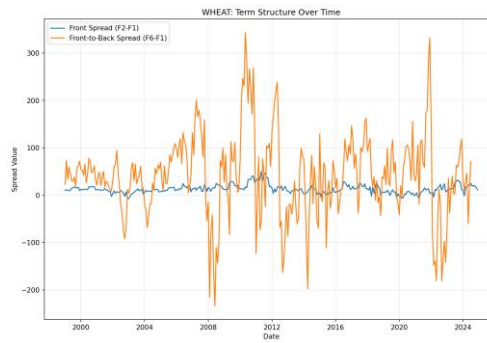
(e) Copper



(f) Corn



(g) Soybeans



(h) Wheat

Note. Monthly series, sample spans 1999M1–2024M12. “Front spread” is $F2 - F1$. “Front-to-back spread” is $F6 - F1$ (or $F5 - F1$ when six-month contracts are sparse). Spreads differences in the futures prices, in currency terms.